International Journal of Multiphase Flow 200 (2026) 105729

Contents lists available at ScienceDirect

International Journal of Multiphase Flow

journal homepage: www.elsevier.com/locate/ijmulflow

Research Paper

A high-order numerical procedure to solve a ten-equation model for high-speed, monodisperse, fluid–particle flows

Victor Boniou [a,*], Rodney O. Fox [a,b], Frédérique Laurent [b]

[a] Department of Chemical and Biological Engineering, Iowa State University, Ames, IA 50011-1098, USA
[b] Laboratoire EM2C, CNRS, CentraleSupélec, Université Paris-Saclay, 3 rue Joliot-Curie, 91192 Gif-sur-Yvette, France

ARTICLE INFO

Keywords:
Fluid–particle flow
Multiscale modeling
Kinetic theory of granular flow
Added mass
Pseudoturbulence
Well-posed two-fluid model

ABSTRACT

In this work, a ten-equation model is presented for the study of high-speed, monodisperse, fluid–particle flows. The model is well-posed for any fluid–particle material density ratio and includes a large range of physical modeling: collisional and frictional pressures in the dense regime, added mass, internal energy and heat transfer between particles, added mass and the fluid. The numerical methods associated with the hyperbolic system are designed to fulfill the main features of a compressible two-phase flow solver: capturing sharp particle fronts, preserving contact discontinuities, and obtaining stability in all flow regimes. This is done by employing a combination of an AUSM$^+$-up scheme for the particle phase, and a HLLC scheme for the fluid phase with limited WENO5 reconstructions. A necessary, but not sufficient, condition for stability is obtained by keeping the discrete consistency between spatial fluxes and non-conservative terms. Test cases combining a high-speed fluid interacting with heavy/light particles are used to demonstrate that the qualitative behavior of the flow dynamics over a wide range of density ratios is captured correctly by the ten-equation model.

1. Introduction

High-speed flows in the presence of a particle phase occur in numerous natural and industrial problems such as volcanic ash plumes (Pelanti and LeVeque, 2006), dust explosions (Benedetto et al., 2010), or rocket propulsion (Davis et al., 2013). The complexity of such multiphase flows stems from the interaction between the shock and the particles, which highly depends on the volume fraction (Zhang et al., 2001). While dilute mixtures can be approximated with experiments and numerical simulations of isolated particles subject to a shock, the presence of neighboring particles cannot be neglected when their volume fraction increases (Ling et al., 2012). In denser mixtures, a particle is not only subject to the incident shock wave, but also to shock reflections and expansion waves coming from the other particles and compaction waves transmitted through the granular phase. When the mixture becomes very dense, the particles are always in contact and the inter-particle interactions become dominant. The proper modeling of granular flows requires to handle these three regimes with an appropriate physical description. Since the pioneering experiments on a gas-shock interaction with particle curtains (Rogue et al., 1998), both numerical and experimental studies have been conducted to give a better description of the shock-induced dispersal of particles. More specifically, experimental studies have been dedicated to the definition of a universal scaling of the spreading rate with respect to material properties, Mach number, and particle concentration (Theofanous et al., 2016; DeMauro et al., 2019; Daniel and Wagner, 2022). These works have been followed up by numerical studies to assess different multiphase flow models (Ling et al., 2012; Houim and Oran, 2016; McGrath et al., 2018). More recently, these studies have been extended to hypersonic flows (Wagner et al., 2023) and underwater shocks (Behrendt et al., 2022).

In this work, we use the novel mesoscale model presented in Fox (2024) for the particles in the form of a kinetic equation. This model accounts for collisional particle flow with added mass and internal energy and is consistent with the kinetic theory of granular flows (KTGF). However, for high-speed systems the thermal energy balance must be carefully modeled (e.g., collisional and frictional heating of the particles), which requires terms that are usually neglected in the KTGF for low-speed flows. For clarity, we consider monodisperse, spherical particles with arbitrary material density and identical physical properties so that only the particle number density function (NDF) for velocity and internal energy is needed in the kinetic model. In most applications, the fluid will either be a gas interacting with high-density solid particles or a liquid interacting with low-density particles (e.g., bubbles). For simplicity, we assume that the particle material density remains constant, which implies that the particle

* Corresponding author.
E-mail addresses: victor.boniou@ifpen.fr (V. Boniou), rofox@iastate.edu (R.O. Fox), frederique.laurent@centralesupelec.fr (F. Laurent).

https://doi.org/10.1016/j.ijmultiphaseflow.2026.105729
Received 28 January 2026; Received in revised form 2 April 2026; Accepted 2 April 2026
Available online 7 April 2026
0301-9322/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

volume is constant. Nonetheless, the extension to allow for the particle material density (and hence the volume) to vary with the fluid pressure is straightforward provided that the particle Stokes number for such cases is small (i.e., bubbly flow). The resulting mesoscale model is valid from the very dilute to the very dense regime (as measured by the particle-phase volume fraction). However, for denser flows the particle–particle interaction terms are dominant and the computational cost of solving the kinetic model directly is prohibitive (and often unnecessary).

A tractable ten-equation model for denser flows can be derived from the mesoscale model by considering the moments of the NDF as the macroscale conserved variables (Fox, 2024). As in the kinetic theory of gases, this process introduces mathematical closures for nonlinear terms based on approximating the shape of the NDF. In the ten-equation model, the particle-phase equations for the conserved moments are coupled with modified Euler equations for the fluid phase, including pseudoturbulent kinetic energy (PTKE) (Shallcross et al., 2020). Due to the high-speed nature of the flow, viscous terms are neglected but can easily be included when needed (Kong and Fox, 2017). Details on the derivation of the forces and fluxes in the monodisperse two-fluid model can be found in prior work (Kong and Fox, 2020; Fox, 2019; Fox et al., 2020; Fox, 2025). The fluid phase is treated as inviscid with coupling terms discussed in Fox et al. (2020), and the stiffened gas equation of state is used for the fluid pressure. Particle velocity moments up to second order (total kinetic energy) are included, which is sufficient for dense collisional flows (Fox, 2014). The macroscopic model then accounts for conservation of mass, momentum and total energy for both phases.

This macroscale model has been extended to polydisperse particles (Fox et al., 2024), phase change (Boniou et al., 2024) and applied to shock–particle-curtain interactions (Boniou and Fox, 2023). However, such a model raises concern in the numerical aspects and the validation on other systems that the gas–solid studies mentioned above, i.e., particles in liquid or bubbly flows. Moreover, an up-to-date derivation of the particle–fluid–particle (pfp) tensor has been recently given (Fox, 2025) which is also included in the solver. This works aim at giving the full picture of the numerical procedure for the hyperbolic solver and how to ensure robustness and accuracy in a large range of applications with perspective for low-Mach applications.

The remainder of this paper is organized as follows. In Section 2, the macroscale kinetic-based model is presented with all the related closures for the coupling terms. In Section 3, the macroscale system is rewritten such that hyperbolic and local terms appear clearly. Then, a robust and accurate hyperbolic solver is presented in Section 4 followed by a brief description of the numerical approach for the additional physics in the form of local source terms in Section 5. Section 6 is devoted to assessment of the hyperbolic solver and 1-D numerical examples including arbitrary material-density ratios ranging from bubbly liquids to gas–particle flows using conditions taken from the literature. Finally, in Section 7, conclusions are drawn and possible extensions to low-Mach-number and combusting flows are discussed.

2. Macroscale kinetic-based model

This section aims to recap the important physical features of the model (Fox, 2024), which are the treatment of added mass, PTKE and internal energy. While added mass affects the averaging definition of the complete system of equations, which lead to a modified seven-equation system (mass, momentum and energy for each phase with an additional transport for volume fraction), the introduction of a more detailed closure of energy dissipation and transfer between phases adds two additional constitutive equations in the system (Fox et al., 2020). Moreover, the framework presented in Boniou et al. (2024) is used in this work with an additional transport equation for the internal energy of the added-mass part, which allows to relax the hypothesis of equal density in the bulk fluid and in the added-mass phase used in Fox et al. (2020) and Boniou and Fox (2023). Finally, the system is updated with the new closure of the pfp tensor presented in Fox (2025).

2.1. Added mass treatment

In our modeling approach (Fox et al., 2020; Cook and Harlow, 1984), added mass is handled by assigning a fraction of the fluid phase surrounding a particle to move with the velocity of the particle. In the context of two-fluid models, we define $\alpha_p^\star = \alpha_p + \alpha_a$ where α_a is the volume fraction of the added-mass phase, while α_p is the particle-phase volume fraction. In the absence of particles, both α_p and α_a are null. It then follows that $\alpha_f^\star = \alpha_f - \alpha_a$ where α_f is the fluid-phase volume fraction. As usual, the volume fractions are non-negative with $\alpha_p + \alpha_f = 1$ and $\alpha_p^\star + \alpha_f^\star = 1$. In this work, the fluid material density is denoted by ρ_f, and the solids material density (assumed to be constant) is ρ_p. Then, the mass of a solid particle is denoted by $m_p = \rho_p V_p$, whereas the particle mass + added mass is denoted by $m_p^\star = \beta m_p$ where β is the added-mass factor leading to the identity $\rho_e \alpha_p^\star = \beta \rho_p \alpha_p$.

In absence of mass transfer, we can invoke the balance equation for the solid mass of particles $\rho_p \alpha_p$ (i.e., without added mass), which is the zero-order moment of the NDF (Kong and Fox, 2020):

$$\partial_t \rho_p \alpha_p + \partial_{\mathbf{x}} \cdot \rho_p \alpha_p \mathbf{u}_p = 0 \tag{1}$$

with ρ_p the particle material density. The kinetic model includes the added mass, denoted by $\rho_a \alpha_a$, so that the total particle mass is $\rho_e \alpha_p^\star = \rho_p \alpha_p + \rho_a \alpha_a$ where ρ_a is the density of the fluid belonging to the added-mass volume. In opposition to the proposed system in Fox et al. (2020) and Boniou and Fox (2023) where $\rho_a = \rho_g$, this work uses pressure equilibrium $p_a = p_g$ instead such that ρ_a is deduced from an equation of state (EOS) with the knowledge of the added-mass temperature T_a. In the previous paper (Boniou et al., 2024), including the phase change in the added-mass system of equations, the ideal-gas EOS is used and the internal energy of the added mass is transported to compute T_a. The same is done here with the stiffened-gas EOS in order to study both gas and liquid flows.

An equivalent expression for ρ_e is

$$\rho_e = \frac{\alpha_p}{\alpha_p^\star}(\rho_p - \rho_a) + \rho_a \tag{2}$$

where $\alpha_p \le \alpha_p^\star$ and ρ_a is obtained from pressure equilibrium $p_a = p_g$. The balance equation for $\rho_e \alpha_p^\star$, derived from the kinetic model is then

$$\partial_t \rho_e \alpha_p^\star + \partial_{\mathbf{x}} \cdot \rho_e \alpha_p^\star \mathbf{u}_p = S_a = \frac{\rho_a}{\tau_a}(\alpha_a^\star - \alpha_a) \tag{3}$$

where $\alpha_a^\star = c_m^\star \alpha_f \alpha_p$ is the equilibrium value of added mass discussed below. In this balance, S_a is the mass-exchange rate between the added-mass phase and the bulk fluid (Fox et al., 2020). The transport equation for the added-mass factor $\beta \ge 1$ is

$$\partial_t \beta + \mathbf{u}_p \cdot \partial_{\mathbf{x}} \beta = \frac{S_a}{\rho_p \alpha_p} = \frac{1}{\tau_a}(\beta^\star - \beta). \tag{4}$$

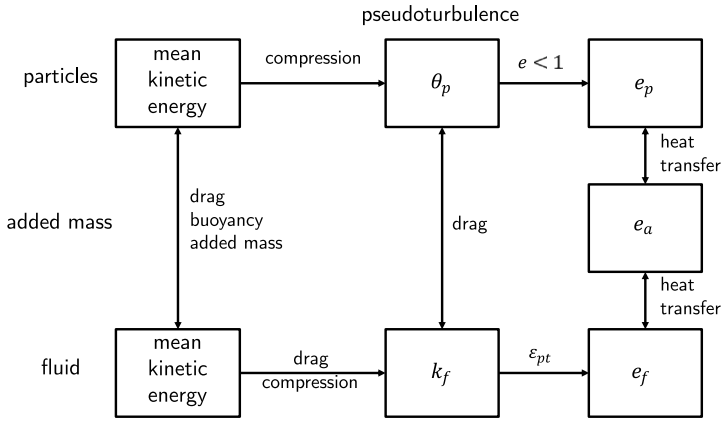


Fig. 1. Energy flow in the fluid–particle, two-fluid model for monodisperse particles. Spatially uncorrelated particle motion is measured by Θ_p, while particle-scale, fluid-phase, fluctuating kinetic energy is measured by k_f. The internal energies are e_p, e_f and e_a where added mass just acts as an insulator between fluid and particles. Inelastic collisions transform Θ_p into an increase in e_p. Pseudoturbulent dissipation ε_{pt} transform k_f into an increase in e_f. At each scale, the phases are coupled through drag and heat transfer, and a drag correlation and the Nusselt number are important inputs to the two-fluid model.

Physically, the added-mass factor is transported with the particle-phase velocity and relaxes towards its local equilibrium value $\beta^\star$. The timescale for relaxation towards equilibrium τ_a is assumed to be proportional to the timescale for single-particle drag τ_p, i.e., $\tau_a \lessapprox \tau_p$. In a spatially homogeneous system, starting with $\beta(0) = 1$, Eq. (4) causes $\beta(t)$ to relax exponentially towards $\beta^\star$, i.e., the added mass is initially zero and relaxes to $m_p^\star - m_p$ with characteristic timescale τ_a. If we write the added-mass volume fraction as $\alpha_a = c_m \alpha_f \alpha_p$ where $0 < c_m(t, \mathbf{x}) < 1$ is the added-mass coefficient with equilibrium value $c_m^\star$, then β and $\beta^\star$ are related to c_m and $c_m^\star$, respectively, by

$$\beta = 1 + \frac{\rho_a}{\rho_p} c_m \alpha_f, \quad \beta^\star = 1 + \frac{\rho_a}{\rho_p} c_m^\star \alpha_f. \tag{5}$$

In this work, the default value of the added-mass coefficient $c_m^\star$ is set to

$$c_m^\star = \frac{1}{2} \min(1 + 2\alpha_p, 2), \tag{6}$$

and hence $\beta^\star$ depends on the material density ratio ρ_f/ρ_p and α_p. For very dilute flows, $c_m^\star = \frac{1}{2}$ as expected for spherical particles (Biesheuvel and Spoelstra, 1989).

2.2. Energy transfer in the system

Physically, the added mass represents the fluid surrounding a particle, also moving with velocity $\mathbf{u}$ (Fox et al., 2020; Cook and Harlow, 1984), but the individual particle's added volumes are strongly coupled together through β, which moves with velocity $\mathbf{u}_p$. Thus, heat and/or mass transfer to (from) the particle phase from (to) the fluid phase passes through the added volume. Hence, an additional energy equation for the added-volume internal energy e_a is also included in the model to account for this energy transfer. In the macroscale model, the fluid- and particle-phase velocities are $\mathbf{u}_f$ and $\mathbf{u}_p$, respectively. The total energies of the fluid and particle phases are denoted as E_f and E_p, respectively. The total energy of the added volume is denoted as E_a. The total energy is made up of kinetic and internal energies. We denote the former for each phase as K_f and K_p (shared between added mass and particle as they are transported at the same speed $\mathbf{u}_p$), and the latter as e_f, e_a and e_p; hence,

$$E_f = K_f + e_f, \quad E_a = K_p + e_a, \quad E_p = K_p + e_p. \tag{7}$$

The kinetic energy is further divided into mean and fluctuating components:

$$K_f = \frac{1}{2} u_f^2 + k_f, \quad K_p = \frac{1}{2} u_p^2 + \frac{3}{2} \Theta_p. \tag{8}$$

Here, $u^2 = \mathbf{u} \cdot \mathbf{u}$, k_f is the PTKE, and Θ_p is the granular temperature. The former represents fluid-velocity fluctuations due to finite-size particles, and the square root of the latter represents the velocity magnitude of individual particles relative to $\mathbf{u}_p$.

In the macroscale model, the total energy of both phases is conserved. For the fluid phase, it is convenient to solve transport equations for E_f and k_f, and to find e_f from Eq. (7). In contrast, for the particle phase, the mesoscale description leads naturally to transport equations for K_p, e_a and e_p, and then Θ_p is found from Eq. (8). In any case, the thermodynamic temperatures T_f, T_a and T_p are found from the internal energies e_f, e_a and e_p, respectively. Hereinafter, for simplicity, we will take the heat capacities as constant such that

$$T_f = \frac{\gamma_f e_f}{C_{p,f}}, \qquad T_a = \frac{\gamma_a e_a}{C_{p,a}}, \qquad T_p = \frac{e_p}{C_{p,p}}. \tag{9}$$

This assumption will be valid as long as the temperatures do not vary too much.

The diagram in Fig. 1 provides a schematic on how kinetic energy is distributed at the macroscale in the fluid–particle flow. The mean kinetic energy is found from the phasic velocities appearing in the momentum balances. Due to their finite size, particles have wakes produced by the slip velocity whose characteristic length scale is proportional to the particle diameter. These wakes interact with individual particles to generate uncorrelated velocity fluctuations measured by the granular temperature (Fox, 2014). This kinetic energy is dissipated to yet smaller scales to modify the internal energy of both phases and added mass as described above. In the absence of body forces (e.g., gravity), the total energy of

the two phases is a conserved quantity, and it must be treated as such in the two-fluid model. In a closed system, the total energy at mechanical equilibrium will reside in the internal energies, such that the thermodynamics temperatures of both phases are equal.

While this model captures velocity fluctuations in the fluid phase, it does not include the kinetic energy due to background turbulence (e.g., from Reynolds averaging (Fox, 2014)). Here, we assume that all the fluid scales are captured by the mesh if particles were absent. If instead a model was required to track the mean fluctuations of the fluid due to unresolved scales which are inherent to the bulk, it would require an additional transport equation for total fluctuations in the fluid κ_f containing both PTKE and mean fluctuations as proposed in Fox (2014) for granular flows.

2.3. Particle–fluid–particle tensor

In this work, the additional terms due to pfp interactions are incorporated in the two-fluid model. The P_{pfp} tensor contains the pairwise interaction between particles through the fluid. When a particle produces disturbances in the fluid flow, these fluctuations are then seen by the neighboring particles, which creates an exchange between phases in the form of an additional stress. While it is often disregarded in the modeling of dispersed flows, the pfp stress can have a significant impact on the phase dynamics when the dispersed phase shows strong inhomogeneity such as particle fronts in mechanical non-equilibrium. The pfp stress must not be mistaken with the short-range pairwise interactions of particles when they collide, which does not produce an exchange between phases but in the particle phase alone. It has been shown in Fox (2019) that the pfp terms help at retrieving hyperbolicity of the two-fluid model. In a recent work (Fox, 2025), these terms have been derived in closed form and also lead to a well-posed two-fluid system, which is here presented including added mass ($\alpha_p \to \alpha_p^\star$ and $\alpha_f \to \alpha_f^\star$). The forces due to spatial gradients can be written in a momentum-exchange term $\mathbf{F}_{pf}$ such that

$$\mathbf{F}_{pf} = -\mathbf{F}_{fp} = \partial_{\mathbf{x}} \cdot \mathbf{P}_{pfp} + \alpha_p^\star \rho_f \boldsymbol{\Gamma}_v \cdot \mathbf{u}_{fp} + \alpha_p^\star \left(\Theta_p \mathbf{I} + \alpha_f^\star \mathbf{R} \right) \cdot \partial_{\mathbf{x}} \rho_f + \alpha_p^\star \rho_f \frac{2}{5} (\partial_{\mathbf{x}} \cdot \mathbf{u}_v) \mathbf{u}_{fp} \tag{10}$$

with the relative velocity $\mathbf{u}_{fp} = \mathbf{u}_f - \mathbf{u}_p$, the volume-average velocity $\mathbf{u}_v = \alpha_p^\star \mathbf{u}_p + \alpha_f^\star \mathbf{u}_f$, and the rate-of-deformation tensor $\boldsymbol{\Gamma}_v = 1/2 \left[(\partial_{\mathbf{x}} \cdot \mathbf{u}_v) + (\partial_{\mathbf{x}} \cdot \mathbf{u}_v)^t \right]$. The tensor $\mathbf{R}$ and the pfp pressure tensor $\mathbf{P}_{pfp}$ are defined as

$$\mathbf{R} = \frac{1}{5} u_{fp}^2 \mathbf{I} + \frac{2}{5} \mathbf{u}_{fp} \otimes \mathbf{u}_{fp}, \quad \mathbf{P}_{pfp} = \rho_f \alpha_p^{\star 2} \mathbf{R}. \tag{11}$$

The corresponding term D_{pf} in the energy equation is then defined as

$$D_{pf} = -D_{fp} = \mathbf{u}_p \cdot \mathbf{F}_{pf} + 2 \alpha_p^\star \Theta_p \left[\mathbf{u}_{pf} \cdot \partial_{\mathbf{x}} \rho_f - \rho_f \partial_{\mathbf{x}} \cdot \mathbf{u}_v - \rho_f \partial_{\mathbf{x}} \cdot \left(\alpha_p^\star \mathbf{u}_{fp} \right) \right]. \tag{12}$$

Also, additional momentum and energy fluxes in the fluid phase are introduced by this formalism:

$$\partial_{\mathbf{x}} \cdot \left[\rho_f \alpha_p^\star \left(\Theta_p \mathbf{I} + \mathbf{R} \right) \right], \quad \partial_{\mathbf{x}} \cdot \left(\alpha_p^\star \rho_f \mathbf{r} \right) \text{ with } \mathbf{r} = \mathbf{R} \cdot \mathbf{u}_p + \Theta_p \left(\mathbf{u}_p + 2 \mathbf{u}_{pf} \right). \tag{13}$$

As shown in Fox (2019), for constant fluid density, the flux eigenvalues can be found explicitly and are always real-valued.

2.4. Governing equations

In conservative form, the particle phase including added mass is governed by mass, momentum, kinetic-energy and internal-energy balances (Fox, 2024):

$$\partial_t \rho_p \alpha_p + \partial_{\mathbf{x}} \cdot \rho_p \alpha_p \mathbf{u}_p = 0, \tag{14}$$

$$\partial_t \rho_a \alpha_a + \partial_{\mathbf{x}} \cdot \rho_a \alpha_a \mathbf{u}_p = S_a, \tag{15}$$

$$\partial_t \rho_e \alpha_p^\star \mathbf{u}_p + \partial_{\mathbf{x}} \cdot \rho_e \alpha_p^\star (\mathbf{u}_p \otimes \mathbf{u}_p + \Theta_p \mathbf{I} + P_p \mathbf{I}) = -\alpha_p^\star \partial_{\mathbf{x}} \hat{p}_f + \mathbf{F}_{fp} - \mathbf{D} + \mathbf{S}_{\mathbf{u}}, \tag{16}$$

$$\partial_t \rho_e \alpha_p^\star K_p + \partial_{\mathbf{x}} \cdot \rho_e \alpha_p^\star (K_p + \Theta_p + P_p) \mathbf{u}_p = -\alpha_p^\star \mathbf{u}_p \cdot \partial_{\mathbf{x}} \hat{p}_f + D_{fp} - D_E - H_p + S_K, \tag{17}$$

$$\partial_t \rho_p \alpha_p e_p + \partial_{\mathbf{x}} \cdot \rho_p \alpha_p e_p \mathbf{u}_p = H_p - H_{pa}, \tag{18}$$

$$\partial_t \rho_a \alpha_a e_a + \partial_{\mathbf{x}} \cdot \rho_a \alpha_a e_a \mathbf{u}_p = H_{pa} - H_{af} + S_e. \tag{19}$$

In this set of equations for particles and added mass, two mass-conservation equations are required to close the system, which are chosen to be $\rho_p \alpha_p$ and $\rho_a \alpha_a$ as in Boniou et al. (2024) while $\rho_e \alpha_p^\star$ was chosen in previous works (Fox et al., 2020; Boniou and Fox, 2023). The density of added mass is closed by transporting the internal energy e_a. The pressure P_p is the total particle pressure due to collisions and friction. The drag is modeled by a momentum exchange term $\mathbf{D}$, and an energy exchange D_E. The term H_p corresponds to the heating of particles due to inelastic collisions, while H_{pa} and H_{af} are the terms for heat transfer from particles to added mass and from added mass to fluid, respectively. The source terms S_a, $\mathbf{S}_{\mathbf{u}}$, S_K and S_e are the mass, momentum, kinetic-energy and internal-energy changes due to added-mass modification. From the momentum and kinetic-energy balances, it can be shown that the pfp pressure tensor does not change Θ_p. Also, the balance equation for the total energy $E_p = K_p + e_p$ can be found by summing Eqs. (17) and (18).

In conservative form, the continuous fluid phase is governed by mass, momentum, total-energy, and PTKE balances (neglecting gravity for simplicity) (Fox et al., 2020):

$$\partial_t \rho_f \alpha_f^\star + \partial_{\mathbf{x}} \cdot \rho_f \alpha_f^\star \mathbf{u}_f = -S_a, \tag{20}$$

$$\partial_t \rho_f \alpha_f^\star \mathbf{u}_f + \partial_{\mathbf{x}} \cdot \left[\rho_f \alpha_f^\star \mathbf{u}_f \otimes \mathbf{u}_f + \rho_f \alpha_p^\star \left(\Theta_p \mathbf{I} + \mathbf{R} \right) \right] = -\alpha_f^\star \partial_{\mathbf{x}} \hat{p}_f + \mathbf{F}_{pf} + \mathbf{D} - \mathbf{S}_{\mathbf{u}}, \tag{21}$$

$$\partial_t \rho_f \alpha_f^\star E_f + \partial_{\mathbf{x}} \cdot (\rho_f \alpha_f^\star \mathbf{u}_f E_f + \alpha_f^\star \mathbf{u}_f \hat{p}_f + \rho_f \alpha_p^\star \mathbf{r}) = -\hat{p}_f \partial_{\mathbf{x}} \cdot \left(\alpha_p^\star \mathbf{u}_p \right) + D_{pf} + D_E + H_{af} - S_E, \tag{22}$$

$$\partial_t \rho_f \alpha_f^\star k_f + \partial_{\mathbf{x}} \cdot \rho_f \alpha_f^\star \mathbf{u}_f k_f + \frac{2}{3} \rho_f \alpha_f^\star k_f \partial_{\mathbf{x}} \cdot \mathbf{u}_f = D_{PT} - \rho_e \alpha_p^\star \frac{C_f}{\tau_p} k_f - S_k. \tag{23}$$

In the last equation for PTKE, D_{PT} is a energy exchange due to drag forces, C_f/τ_p controls the dissipation rate of PTKE with τ_p the drag timescale, and S_k is the PTKE change due to added mass. In the equation for the total energy, S_E is the energy change due to added mass such that $S_E = S_K + S_e$. All the closures for the pressure and source terms in the equations presented above are provided in Section 2.5.

Comparing the total-energy balances, it can be easily seen that the sum of the total energies of the two phases is conserved. The same is true for the mass and momentum balances. In the absence of the frictional pressure, the mathematical properties of the macroscale model have been investigated in Fox et al. (2020) where it is shown that the model is well-posed. Note that when the frictional pressure is present ($\alpha_p > \alpha_{max}$), it will determine the speed of sound in the particle phase, independent of the fluid phase.

2.5. Physical closures

In the ten-equation system, equations are coupled through transport and source terms. This section presents the choice of physical closures for these terms.

2.5.1. Pressure terms

The particle-phase pressure $P_p = P_c + P_{fr}$ arises due to particle–particle interactions in the KTGF based on the Boltzmann–Enskog equation for finite-size particles (i.e., excluded volume) (Lun et al., 1984; Gidaspow, 1994; Garzó et al., 2007; Marchisio and Fox, 2013). Note that because P_p does not depend on $\mathbf{u}$, the pressure terms can alternatively be treated as acceleration terms. The model for P_p depends on the velocity moments up to second order (Fox, 2019), and is the sum of a collisional component:

$$P_c = 2(1 + e_c)\alpha_p g_0 \Theta_p \quad \text{with} \quad g_0 = \frac{1 + \alpha_f}{2\alpha_f^3} \tag{24}$$

where e_c is the coefficient of restitution; and a frictional (or configuration (Saurel et al., 2010)) component (Johnson and Jackson, 1987), modeled here as

$$P_{fr} = \frac{p_{fr} \alpha_p g_0}{\rho_e \alpha_p^\star} h_{fr} \quad \text{with} \quad h_{fr} = \frac{1}{2} \left[1 + \tanh \left(\frac{\alpha_p - \alpha_{max}}{\Delta_{fr}} \right) \right] \tag{25}$$

where α_{max} is the close-packed limit and Δ_{fr} controls the sharpness of the transition to α_{max}. The parameter p_{fr} fixes the speed of sound in the particle phase above the close-packed limit. The functional form of h_{fr} causes P_{fr} to rapidly increase from zero near $\alpha_p = \alpha_{max} \approx 0.63$. The expression for the frictional pressure is material dependent and is also referred to as granular or configuration pressure (Saurel et al., 2010). Consistent with assuming that ρ_p is constant, in this work we neglect the irreversible part of the frictional pressure, which requires a dedicated EOS (Saurel et al., 2010). In any case, the granular EOS for P_{fr} must ensure that the frictional pressure increases monotonely with volume fraction, i.e., $d\rho_e \alpha_p^\star P_{fr}/d\alpha_p > 0$. In general, if $\alpha_p < 0.001$ (dilute regime), then the particle-phase pressure P_p is negligible. Finally, note that added mass plays no role in the granular EOS for frictional pressure $\rho_e \alpha_p^\star P_{fr}$ because the particles are in contact. Other forms for P_c that account for velocity correlations between colliding particles are also available (Jenkins and Larcher, 2023). Here, for clarity, when the particle pressure depends on both α_p and Θ_p, we refer to it as collisional; otherwise, when it depends only on α_p, we refer to it as frictional. In any case, although it is negligible for very dilute flows, P_p controls the particle-phase speed of sound for dense flows.

The fluid pressure is found from an equation of state. In this work, we will use the stiffened-gas law:

$$p_f = (\gamma_f - 1)\, \rho_f \left(E_f - K_f \right) - \gamma_f p_{\infty,f} = (\gamma_f - 1)\, \rho_f e_f - \gamma_f p_{\infty,f} \tag{26}$$

where γ_f is the heat-capacity ratio. For an ideal gas, the reference pressure $p_{\infty,f}$ is set to 0, leading to the well-known ideal-gas law. In liquids, $p_{\infty,f} > 0$ is used to control the speed of sound. Then, the full fluid pressure is obtained by adding the PTKE contribution

$$\hat{p}_f = p_f + \frac{2}{3} \rho_f k_f. \tag{27}$$

For the added-mass fluid, the pressure can also be defined from

$$p_a = (\gamma_a - 1)\, \rho_a e_a - \gamma_a p_{\infty,a}, \tag{28}$$

and is assumed to be equal to p_f, which fixes the value of ρ_a given e_a.

2.5.2. Drag terms

The terms $\mathbf{D}$, D_E and D_{PT} represent drag exchange:

$$\mathbf{D} = \frac{\rho_e \alpha_p^\star}{\tau_p} \mathbf{u}_{pf}, \tag{29}$$

$$D_E = \frac{\rho_e \alpha_p^\star}{\tau_p} [3a\Theta_p - 2(1 - a)k_f + \mathbf{u}_{pf} \cdot \mathbf{u}_p], \tag{30}$$

$$D_{PT} = \frac{\rho_e \alpha_p^\star}{\tau_p} [u_{pf}^2 + 3a\Theta_p - 2(1 - a)k_f]. \tag{31}$$

Physically, $D_{PT} \geq 0$ represents kinetic energy lost from the particle phase to produce PTKE in the fluid phase. This kinetic energy comes from both the mean (u_p^2) and fluctuating (Θ_p) components. There is kinetic-energy transfer to the fluid from particles due to drag when $D_E > 0$ or vice versa when $D_E < 0$.

The drag timescale τ_p depends on α_p and $\mathbf{u}_{pf}$:

$$\tau_p = \frac{4\rho_e d_p^2}{3\mu_f C_D Re_p} = \frac{4}{3}\frac{\alpha_f}{\alpha_f^\star}\beta\frac{\rho_p d_p^2}{\mu_f C_D Re_p}, \tag{32}$$

and the particle Reynolds number is

$$Re_p = \frac{\rho_f d_p v_{pf}}{\mu_f}. \tag{33}$$

In the definition of Re_p, the slip velocity v_{pf} is defined in the absence of added mass (i.e., $\beta = 1$, $\alpha_f^\star = \alpha_f$). As shown in Fox et al. (2020), the slip velocities with and without added mass are related by $\alpha_f \mathbf{v}_{pf} = \alpha_f^\star \mathbf{u}_{pf}$. The slip velocity with added mass is larger because part of the fluid moves with the particle velocity $\mathbf{u}_p$. However, drag models in the literature are based on v_{pf}, so that τ_p in Eq. (32) is multiplied by $\alpha_f/\alpha_f^\star$ to recover the correct drag force. Here, d_p is the particle diameter and C_D is a drag coefficient that depends on the particle Reynolds number and other dimensionless parameters such as the Mach number and volume fraction. For Stokes flow, which is valid for very small particles and $\alpha_f = 1$, $C_D Re_p = 24$. In this work, we employ the correlation for an isolated particle (Naumann and Schiller, 1935) multiplied by the Richardson–Zaki empirical relation (Richardson and Zaki, 1997) to account for collective (swarm) effects:

$$C_D = \frac{24}{Re_p}\left(1 + 0.15 Re_p^{0.687}\right)\alpha_g^{-2.65}. \tag{34}$$

For energy-related drag terms, the parameter a depends on the material densities, and is modeled as

$$\frac{1-a}{1-a_{min}} = \frac{\rho_f}{\rho_p K + \rho_f} \tag{35}$$

where $a_{min} = 0.5$ determines the steady-state ratio Θ_p/k_f for gas bubbles in a liquid (i.e., $\rho_p/\rho_f \ll 1$), and $K = 0.06$ fixes the steady-state ratio for $\rho_f = \rho_p$ (Tavanashad et al., 2019). Heavy particles in a gas correspond to $a \lessapprox 1$.

Finally, the PTKE parameter C_f in Eq. (23) controls the rate of dissipation of PTKE into internal energy in the fluid phase (Shallcross et al., 2020). Here, we will use the correlation from particle-resolved DNS with frozen particles (Mehrabadi et al., 2015):

$$C_f^{-1} = \alpha_p[1 + 1.25\alpha_f^3 \exp(-\alpha_p \alpha_f^{1/2} Re_p^{1/2})]. \tag{36}$$

At a steady state with constant u_{pf}, we then find

$$\frac{k_f}{u_{pf}^2} \to C_f^{-1}, \quad \Theta_p \to \frac{2(1-a)}{3a}k_f. \tag{37}$$

The correlation in (36) is valid for uniform α_p. When k_f is transported from a uniform region to a very dilute region (e.g., behind a particle curtain), C_f from (36) can be much too large. Possible alternatives are to replace, in dilute regions, C_f/τ_p by $C_f^* k_f^{1/2}/d_p$ where C_f^* is a constant of order one, or to solve a transport equation for pseudoturbulence frequency ω_{pt} (Shallcross et al., 2020; Fox, 2021). Recently, a k–ε model has been proposed to better capture the PTKE dissipation downstream of a shock interacting with a particle curtain (Sridhar et al., 2025).

2.5.3. Added mass

In the model for added mass, the added mass moving with the particles changes to adapt to the local value of α_p at the rate S_a. This rate does not include possible mass transfer from/to the particles to/from the fluid (e.g., phase change), which is neglected in this work for clarity but can be found in a related work (Boniou et al., 2024). Thus, when S_a is positive, fluid is added to the particle wake with density $\hat{\rho} = \rho_f$, while the opposite occurs when S_a is negative with density $\hat{\rho} = \rho_a$. Note that if the fluid temperature and composition is the same, then $\rho_f = \rho_a$. Such changes will result in changes in the mass, momentum and kinetic energy of each phase in the macroscale model corresponding to S_a, $\mathbf{S}_\mathbf{u}$, S_K, S_e, S_E and S_k. The added-mass source term is defined as

$$S_a = \frac{\hat{\rho}}{\tau_a}\left(\alpha_a^\star - \alpha_a\right) \tag{38}$$

with $\alpha_a^\star = c_m^\star \alpha_f \alpha_p$ and $\tau_a = C_a \tau_p$ a timescale proportional to the drag timescale as discussed in Section 2.1. In this work, $C_a = 1$. Then the other added-mass source terms are found from

$$\mathbf{S}_\mathbf{u} = \max(S_a, 0)\,\mathbf{u}_f + \min(S_a, 0)\,\mathbf{u}_p, \tag{39}$$

$$S_K = \max(S_a, 0)\,K_f + \min(S_a, 0)\,K_p, \tag{40}$$

and

$$S_e = \max(S_a, 0)\,e_f + \min(S_a, 0)\,e_p. \tag{41}$$

As shown below, these source terms ensure conservation of mass, internal energy, momentum, and kinetic energy for the sum of the two phases. Moreover, for the PTKE and total-energy source terms needed for the fluid phase, we have

$$S_k = \max(S_a, 0)\,k_f + \min(S_a, 0)\,\frac{3}{2}\Theta_p, \tag{42}$$

and

$$S_E = \max(S_a, 0)\,E_f + \min(S_a, 0)\,E_p, \tag{43}$$

such that $S_E = S_K + S_e$ and $S_K - S_k = \max(S_a, 0)\frac{1}{2}u_f^2 + \min(S_a, 0)\frac{1}{2}u_p^2$ as expected. All of these processes occur with constant solid particle mass $\rho_p\alpha_p$, i.e., only the added mass composed of fluid changes. In other words, at the macroscale, the added-mass factor β and $\rho_e\alpha_p^\star$ change when S_a is nonzero, but not $\rho_p\alpha_p$. The balance equation for β in Eq. (4) results from this constraint.

2.5.4. Heat transfer

The terms H_{pa} and H_{af} in Eqs. (18), (19) and (22) account for exchange of internal energy in each phase due to the temperature difference. Using the direct relations given by Eq. (9), we can define these terms by

$$H_{pa} = \frac{6\alpha_p\lambda_a Nu_p}{d_p^2}(T_p - T_a), \quad H_{af} = \frac{6\alpha_p^\star\lambda_f Nu_a}{d_a^2}(T_a - T_f) \tag{44}$$

where λ_a and λ_f are the added-mass and fluid-phase thermal conductivities, respectively, and the diameter of the added volume is defined by $d_a = d_p(\alpha_p^\star/\alpha_p)^{1/3}$. In the definition of H_{af}, $\alpha_p^\star$ is used instead of α_a as proposed in the previous work (Boniou et al., 2024) to better represent the exchange surface between added mass and fluid phases. The Gunn correlation (Gunn, 1978) for the added-mass Nusselt number (Nu_a) is defined by

$$Nu_a = (7 - 10\alpha_f^\star + 5\alpha_f^{\star 2})(1 + 0.7Re_p^{0.2}Pr_f^{1/3}) + (1.33 - 2.4\alpha_f^\star + 1.2\alpha_f^{2\star})Re_a^{0.7}Pr_f^{1/3} \tag{45}$$

with the Prandtl number $Pr_f = \rho_f C_{p,f}\nu_f/\lambda_f$ and the added-mass Reynolds number is $Re_a = \rho_f d_a v_{pf}/\mu_f$. The Nusselt number for the particles is defined by

$$Nu_p = (7 - 10\alpha_f^\star + 5\alpha_f^{\star 2})\frac{d_a + d_p}{d_a - d_p}. \tag{46}$$

In most cases without phase change $T_a \approx T_f$, leading to $\rho_a \approx \rho_f$. However, the relaxation times are slightly different leading to small deviations as will be shown in the results section.

2.5.5. Collisions and friction

In general, the inelastic BGK model can be written as (Passalacqua et al., 2011)

$$C(\mathbf{u}, e) = \frac{\rho_e\alpha_p^\star f_G(\mathbf{u}, e) - f(\mathbf{u}, e)}{\tau_c} \tag{47}$$

where f_G is a Gaussian distribution with the same mean velocity as f (i.e., $\int \mathbf{u} f_G \,\mathrm{d}\mathbf{u}\,\mathrm{d}e = \mathbf{u}_p$) and τ_c is the collision time that depends on α_p and Θ_p:

$$\tau_c = \frac{d_p\sqrt{\pi}}{12\alpha_p g_0 \Theta_p^{1/2}}. \tag{48}$$

For inelastic collisions ($e_c < 1$), the covariance matrix of f_G depends on $\omega = (1 + e_c)/2$ (Marchisio and Fox, 2013), as well as its mean value in e, which is $\int e f_G \,\mathrm{d}\mathbf{u}\,\mathrm{d}e = e_p + (1 - e_c^2)\Theta_p$. In this work, the second-order cross moment in f_G is set to $\int \mathbf{u}e f_G \,\mathrm{d}\mathbf{u}\,\mathrm{d}e = \mathbf{u}_p \int e f_G \,\mathrm{d}\mathbf{u}\,\mathrm{d}e$ (i.e., during a collision $\mathbf{u}$ and e are uncorrelated).

The frictional source term F is applicable to dense flows with sustained contacts. Physically, the sole purpose of F is to drive the granular temperature Θ_p to a negligible value when $\alpha_p > \alpha_{max}$, thereby producing particle-phase internal energy e_p such that the total energy E_p is conserved. At the same time, all particle velocities are pushed towards $\mathbf{u}_p$, thereby conserving momentum. To satisfy these constraints, the frictional model used in this work is

$$F(\mathbf{u}, e) = \frac{\rho_e\alpha_p^\star\delta(\mathbf{u} - \mathbf{u}_p)\delta(e - e_p - \Theta_p) - f(\mathbf{u}, e)}{\tau_{fr}} \tag{49}$$

where the delta function $\delta(\mathbf{u} - \mathbf{u}_p)$ ensures momentum conservation and $\delta(e - e_p - \Theta_p)$ ensures conservation of E_p. τ_{fr} is a timescale that depends on τ_c and $\partial_{\mathbf{x}} \cdot \mathbf{u}_p$:

$$\tau_{fr} = \frac{c_{fr}}{\max(|\partial_{\mathbf{x}} \cdot \mathbf{u}_p|, 1/\tau_c)h_{fr}}. \tag{50}$$

Here, $c_{fr} = 0.01$ and h_{fr} control the frictional timescale above close packing. The τ_c-dependence makes τ_{fr} very small when $\alpha_p > \alpha_{max}$. The $|\partial_{\mathbf{x}} \cdot \mathbf{u}_p|$-dependence attempts to capture the effect of relative particle motion under such conditions (i.e., in the absence of deformation there is no friction). The frictional contribution due to the transfer of particle-phase kinetic energy to internal energy is

$$-\partial_t\rho_e\alpha_p^\star K_p\Big|_{fr} = \rho_e\alpha_p^\star\frac{1}{\tau_{fr}}\Theta_p, \tag{51}$$

which drives Θ_p quickly to zero above close packing. The collisional contribution to the particle-phase heating term is

$$-\partial_t\rho_e\alpha_p^\star K_p\Big|_{c} = \rho_e\alpha_p^\star\frac{(1 - e_c^2)}{\tau_c}\Theta_p. \tag{52}$$

The particle-phase heating rate in Eq. (18) is thus

$$H_p = \rho_e\alpha_p^\star\left[\frac{1}{\tau_c}(1 - e_c^2) + \frac{1}{\tau_{fr}}\right]\Theta_p \tag{53}$$

where the collision time τ_c is defined in Eq. (48) and the frictional time scale τ_{fr} is defined in Eq. (50).

Table 1
Ten-equation macroscale model for monodisperse fluid–particle flow with added mass, internal energy for particle and added mass, and PTKE. Gravity $\mathbf{g}$ is included in the model for completeness.

Mass balances:

$$\partial_t \rho_p \alpha_p + \partial_{\mathbf{x}} \cdot \rho_p \alpha_p \mathbf{u}_p = 0$$

$$\partial_t \rho_a \alpha_a + \partial_{\mathbf{x}} \cdot \rho_a \alpha_a \mathbf{u}_p = S_a$$

$$\partial_t \rho_f \alpha_f^\star + \partial_{\mathbf{x}} \cdot \rho_f \alpha_f^\star \mathbf{u}_f = -S_a$$

Momentum balances:

$$\partial_t \rho_f \alpha_f^\star \mathbf{u}_f + \partial_{\mathbf{x}} \cdot \left[\rho_f \alpha_f^\star \mathbf{u}_f \otimes \mathbf{u}_f + \rho_f \alpha_p^\star \left(\Theta_p \mathbf{I} + \mathbf{R}\right)\right] = -\alpha_f^\star \partial_{\mathbf{x}} \hat{p}_f - \mathbf{F}_{pf} + \tfrac{\rho_e \alpha_p^\star}{\tau_p} \mathbf{u}_{pf} - \mathbf{S}_{\mathbf{u}} + \rho_f \alpha_f^\star \mathbf{g}$$

$$\partial_t \rho_e \alpha_p^\star \mathbf{u}_p + \partial_{\mathbf{x}} \cdot \rho_e \alpha_p^\star (\mathbf{u}_p \otimes \mathbf{u}_p + \Theta_p \mathbf{I} + P_p \mathbf{I}) = -\alpha_p^\star \partial_{\mathbf{x}} \hat{p}_f + \mathbf{F}_{pf} - \tfrac{\rho_e \alpha_p^\star}{\tau_p} \mathbf{u}_{pf} + \mathbf{S}_{\mathbf{u}} + \rho_e \alpha_p^\star \mathbf{g}$$

Bulk fluid-phase energy balances (total and PTKE):

$$\partial_t \rho_f \alpha_f^\star E_f + \partial_{\mathbf{x}} \cdot (\rho_f \alpha_f^\star \mathbf{u}_f E_f + \alpha_f^\star \mathbf{u}_f \hat{p}_f + \rho_f \alpha_p^\star \mathbf{r}) = -\hat{p}_f \partial_{\mathbf{x}} \cdot \left(\alpha_p^\star \mathbf{u}_p\right) + D_{fp}$$
$$+ \tfrac{\rho_e \alpha_p^\star}{\tau_p} [3a\Theta_p - 2(1-a)k_f + \mathbf{u}_{pf} \cdot \mathbf{u}_p] + H_{af} - S_E + \rho_f \alpha_f^\star \mathbf{u}_f \cdot \mathbf{g}$$

$$\partial_t \rho_f \alpha_f^\star k_f + \partial_{\mathbf{x}} \cdot \rho_f \alpha_f^\star \mathbf{u}_f k_f = -\tfrac{2}{3} \rho_f \alpha_f^\star k_f \partial_{\mathbf{x}} \cdot \mathbf{u}_f + \tfrac{\rho_e \alpha_p^\star}{\tau_p} [3a\Theta_p - 2(1-a)k_f + u_{pf}^2 - C_f k_f] - S_k$$

Particle-phase (including added mass) energy balances (kinetic and internal):

$$\partial_t \rho_e \alpha_p^\star K_p + \partial_{\mathbf{x}} \cdot \rho_e \alpha_p^\star (K_p + \Theta_p + P_p) \mathbf{u}_p = -\alpha_p^\star \mathbf{u}_p \cdot \partial_{\mathbf{x}} \hat{p}_f + D_{pf}$$
$$- \tfrac{\rho_e \alpha_p^\star}{\tau_p} [3a\Theta_p - 2(1-a)k_f + \mathbf{u}_{pf} \cdot \mathbf{u}_p] - H_p + S_K + \rho_e \alpha_p^\star \mathbf{u}_p \cdot \mathbf{g}$$

$$\partial_t \rho_p \alpha_p e_p + \partial_{\mathbf{x}} \cdot \rho_p \alpha_p e_p \mathbf{u}_p = H_p - H_{pa}$$

$$\partial_t \rho_a \alpha_a e_a + \partial_{\mathbf{x}} \cdot \rho_a \alpha_a e_a \mathbf{u}_p = H_{pa} - H_{af} + S_e$$

2.6. Determination of primitive variables from conserved variables

Finally, the procedure to perform the mapping from conserved variables to primitive variables related to mass is described based on pressure equilibrium $p_a = p_f$. This is analogous to performing a vapor–liquid equilibrium in the context of thermodynamically defined two-phase flows. From the transported variables $(X_1, X_2, X_3) = (\alpha_p \rho_p, \alpha_a \rho_a, \alpha_f^\star \rho_f)$, the primitive variables $(\alpha_a, \rho_a, \rho_g)$ can be retrieved from the following system using basic relations and equation of states defined in Eqs. (26) and (28):

$$\begin{cases} \alpha_a \rho_a = X_2 \\ (1 - \frac{X_1}{\rho_p} - \alpha_a)\rho_f = X_3 \\ (\gamma_a - 1)\rho_a e_a - \gamma_a p_{\infty,a} = (\gamma_f - 1)\rho_f e_f - \gamma_f p_{\infty,f} \end{cases}. \tag{54}$$

This results in solving the following quadratic equation in α_a:

$$\tilde{\kappa}\alpha_a^2 - (\hat{\kappa} + \kappa + \tilde{\kappa})\alpha_f \alpha_a + \hat{\kappa}\alpha_f^2 = 0 \tag{55}$$

with the parameters defined as

$$\alpha_p = \frac{X_1}{\rho_p},\ \alpha_f = 1 - \alpha_p,\ \kappa = \frac{(\gamma_f - 1)e_f}{(\gamma_a - 1)e_a},\ \hat{\kappa} = \frac{X_2}{X_3},\ \tilde{\kappa} = \frac{\alpha_f \Delta\rho_\infty}{X_3},\ \Delta\rho_\infty = \frac{\gamma_a p_{\infty,a} - \gamma_f p_{\infty,f}}{(\gamma_a - 1)e_a}. \tag{56}$$

Once α_a is retrieved from Eq. (55), the other primitive variables are

$$\rho_a = \frac{X_2}{\alpha_a},\quad \alpha_p^\star = \alpha_p + \alpha_a,\quad \alpha_f^\star = 1 - \alpha_p^\star,\quad \rho_f = \frac{X_3}{\alpha_f^\star}. \tag{57}$$

Note that if the bulk and the added mass have the same fluid properties ($\gamma_a = \gamma_f$ and $p_{\infty,a} = p_{\infty,f}$) or are following an ideal-gas law ($p_{\infty,a} = p_{\infty,f} = 0$), then $\tilde{\kappa}$ is null, and the quadratic relation Eq. (55) reduces to the linear relation given in Boniou et al. (2024):

$$\alpha_a = \frac{\hat{\kappa}}{\hat{\kappa} + \kappa}\alpha_f. \tag{58}$$

The ten-equation macroscale model for fluid–particle flow with added mass, internal energy, and PTKE is summarized in Table 1 with the modeling described in Section 2.5. For specific fluid and solid phases, the thermodynamic properties must be specified for each application. For example, it may be necessary to account for the temperature dependence of the heat capacities when defining the internal energies, and the fluid-phase viscosity. For the shock–particle-curtain cases, the definition of the drag coefficient C_D may need to depend on the Mach number (Capecelatro, 2022; Boniou and Fox, 2023).

3. Solver overview

The solution procedure for the coupled fluid–particle system of equations is now detailed. For clarity, the ten equations summarized in Table 1 are rewritten in 1-D as

$$\partial_t \mathbf{Y} + \partial_x \boldsymbol{f}(\mathbf{Y}) = \boldsymbol{h}(\mathbf{Y}) + \boldsymbol{s}(\mathbf{Y}). \tag{59}$$

The conservative-variable and flux vectors are defined, respectively, as

$$\mathbf{Y} = \begin{bmatrix} \rho_p\alpha_p \\ \rho_a\alpha_a \\ \rho_f\alpha_f^\star \\ \rho_f\alpha_f^\star u_f \\ \rho_e\alpha_p^\star u_p \\ \rho_f\alpha_f^\star E_f \\ \rho_e\alpha_p^\star K_p \\ \rho_f\alpha_f^\star k_f \\ \rho_p\alpha_p e_p \\ \rho_a\alpha_a e_a \end{bmatrix}, \quad \boldsymbol{f}(\mathbf{Y}) = \begin{bmatrix} \rho_p\alpha_p u_p \\ \rho_a\alpha_a u_p \\ \rho_f\alpha_f^\star u_f \\ \rho_f\alpha_f^\star u_f^2 + \rho_f\alpha_p^\star\left(\Theta_p\mathbf{I} + \mathbf{R}\right) \\ \rho_e\alpha_p^\star(u_p^2 + \Theta_p + P_p) \\ \alpha_f^\star(\rho_f E_f + \hat{p}_f)u_f + \rho_f\alpha_p^\star\mathbf{r} \\ \rho_e\alpha_p^\star(K_p + \Theta_p + P_p)u_p \\ \rho_f\alpha_f^\star k_f u_f \\ \rho_p\alpha_p e_p u_p \\ \rho_a\alpha_a e_a u_p \end{bmatrix}. \tag{60}$$

The non-conservative source-term and the local source-term vectors are defined, respectively, as

$$\boldsymbol{h}(\mathbf{Y}) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -\alpha_f^\star\partial_x\hat{p}_f + F_{pf} \\ -\alpha_p^\star\partial_x\hat{p}_f - F_{pf} \\ -\hat{p}_f\partial_x(\alpha_p^\star u_p) + D_{pf} \\ -\alpha_p^\star u_p\partial_x\hat{p}_f - D_{pf} \\ -\frac{2}{3}\rho_f\alpha_f^\star k_f\partial_x u_f \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{s}(\mathbf{Y}) = \begin{bmatrix} 0 \\ S_a \\ -S_a \\ D - S_{\mathbf{u}} \\ -D + S_{\mathbf{u}} \\ D_E + H_{af} - S_E \\ -D_E - H_p + S_K \\ D_{PT} - \rho_e\alpha_p^\star\frac{C_f}{\tau_p}k_f - S_k \\ H_p - H_{pa} \\ H_{pa} - H_{af} + S_e \end{bmatrix} \tag{61}$$

where D is the fluid-drag term in the 1-D momentum balances. For simplicity, gravity is not included in the local source terms as it is negligible in the dynamics of high-speed flows.

The separation between non-conservative and local source terms is important as they have different natures. Non-conservative terms are the root of several mathematical and numerical challenges when solving two-fluid systems. Those terms involve derivatives of $\mathbf{Y}$, which play a role in the eigenvalues of the system (Lhuillier et al., 2013). Also, it is of primary importance to keep a discrete consistency between $\boldsymbol{f}(\mathbf{Y})$ and $\boldsymbol{h}(\mathbf{Y})$ to prevent pressure disturbances (Saurel and Abgrall, 1999). In this work, the pressure non-disturbing condition is obtained by regrouping pressure terms as in Houim and Oran (2016). Then, for the fluid momentum and energy Eq. (21) and (22), respectively, the pressure terms in the fluxes have been regrouped with the non-conservative terms as

$$\alpha_p^\star\partial_x\hat{p}_f - \partial_x\hat{p}_f = -\alpha_f^\star\partial_x\hat{p}_f, \quad \alpha_p^\star u_p\partial_x\hat{p}_f - \partial_x(\alpha_p^\star\hat{p}_f u_p) = -\hat{p}_f\partial_x(\alpha_p^\star u_p). \tag{62}$$

The consequence of this rewriting is a pressure-less fluid-momentum flux, which is discussed in Section 4.3, and a loss of discrete conservation of total energy of both phases as non-conservative terms are no longer compensating. The other non-conservative terms arising from the particle–fluid–particle tensor $\mathbf{F}_{pf}$ and D_{pf} are handled using consistent face reconstructions detailed in Section 4.4

The local source term $\boldsymbol{s}(\mathbf{Y})$ can also be challenging to solve numerically because of the potential temporal stiffness (Pelanti and LeVeque, 2006). For this reason, the system is split into a hyperbolic operator $\mathcal{H}_{\Delta t}$ containing the conservative fluxes $\partial_x\boldsymbol{f}(\mathbf{Y})$, the non-conservative source term $\boldsymbol{h}(\mathbf{Y})$, and the source-term operator $\mathcal{S}_{\Delta t}$ containing the local source term $\boldsymbol{s}(\mathbf{Y})$ such that integration over a time step Δt leads to

$$\mathbf{Y}^{n+1} = \mathcal{H}_{\Delta t/2}\mathcal{S}_{\Delta t}\mathcal{H}_{\Delta t/2}\mathbf{Y}^n. \tag{63}$$

In the above time integration, a Strang splitting is used to reduce operator-splitting errors.

4. Hyperbolic solver

The hyperbolic step $\mathcal{H}_{\Delta t}$ is obtained by applying the Godunov scheme to Eq. (59) without source terms (Saurel and Abgrall, 1999):

$$\mathbf{Y}_i^{n+1} = \mathbf{Y}_i^n - \frac{\Delta t}{\Delta x}(\boldsymbol{f}_{i+\frac{1}{2}} - \boldsymbol{f}_{i-\frac{1}{2}}) + \Delta t\,\boldsymbol{h}_i \tag{64}$$

where n and $n+1$ superscripts stand for quantities at time t and $t+\Delta t$, respectively. The subscripts i, $i-\frac{1}{2}$ and $i+\frac{1}{2}$ are the volume average, left-, and right-face, respectively, reconstruction of a quantity defined in the ith computational cell with Δx the cell size.

The computation of spatial fluxes can be tedious because of the discontinuous nature of shocks in the fluid phase and sharp fronts in the particle phase. In Fox et al. (2020) a classic HLL scheme was employed to solve for the full system of equations with first-order reconstructions of quantities. While the extension of this scheme to the present nine-equation model is straightforward, and very robust, it introduces excessive diffusion of discontinuities, which can be prohibitive for the accurate simulation of strong shocks or thin particle layers. Also the discretization of non-conservative terms did not allow for mechanical equilibrium.

An improvement can be obtained by considering higher-order reconstructions to better capture discontinuities. In Vié et al. (2015), a MUSCL reconstruction with a second-order Runge–Kutta integration in time is applied to a moment method, leading to a clear reduction of numerical diffusion. However, this improvement does not cure the severe diffusion of the particle front in our case because of the large disparity in the magnitude of the fluid and particle eigenvalues. The particle eigenvalues are usually smaller by several orders of magnitude. When the volume fraction approaches the packing limit, the frictional pressure grows very fast, and the resulting particle eigenvalues are higher. This observation is the main motivation for splitting the system into a particle subsystem and a fluid subsystem treated with different flux reconstructions. Interested readers can refer to Benjelloun and Ghidaglia (2021) for a discussion on the sound speed definition in two-phase flows.

In this work, the approach proposed in Houim and Oran (2016) is used. It consists of a combination of the AUSM$^+$-up scheme (Liou, 2006) for the particle subsystem, and the HLLC scheme (Toro et al., 1994) for the fluid sub-system. Specific adjustments are required to obtain a well-posed scheme in the presence of added mass and pfp pressure. As for the HLL scheme used in Fox et al. (2020), the numerical fluxes using upwind reconstruction are first order in space, but with enhanced particle front capturing due to the AUSM$^+$-up scheme and contact discontinuity capturing in the fluid thanks to the HLLC scheme. A higher-order extension of the scheme is also provided in this section by using higher-order reconstructions and a multistage integration in time based on the Runge–Kutta method to retrieve a TVD scheme (Gottlieb et al., 2001). To achieve the extension to higher order, special care is devoted to the discretization of face quantities and non-conservative terms to retrieve the non-disturbing properties of the scheme.

In the following, the computation of face fluxes for each subsystem is detailed and the non-conservative source term is finally computed accordingly. Note that the extension to higher dimensions is straightforward employing directional splitting. The only issue that needs to be tackled is the potential carbuncle instability produced by the HLLC scheme when a shock is aligned with a Cartesian mesh direction. This can be solved by using a rotational algorithm as in Huang et al. (2011).

4.1. Reconstruction of left and right states

One major ingredient in the computation of fluxes is the face reconstruction of conservative variables. The superscripts L and R refer to the reconstruction of a quantity at the face $i+\frac{1}{2}$ and stand for left and right states, respectively. For a quantity $\boldsymbol{\Phi}$ at face $i+\frac{1}{2}$ and its left-biased high-order approximation $\boldsymbol{\Phi}^{HO,L}_{i+\frac{1}{2}}$, the left reconstruction $\boldsymbol{\Phi}^{L}_{i+\frac{1}{2}}$ is expressed as

$$\boldsymbol{\Phi}^{L}_{i+\frac{1}{2}} = \boldsymbol{\Phi}_i + \frac{1}{2}(\boldsymbol{\Phi}_i - \boldsymbol{\Phi}_{i-1})\, r_i^{TVD} \tag{65}$$

with the slope limiter r_i^{TVD} defined as in Houim and Kuo (2011)

$$r_i^{TVD} = \max[0, \min(2, 2r_i, 2r_i^{HO})], \quad r_i = \frac{\boldsymbol{\Phi}_{i+1} - \boldsymbol{\Phi}_i}{\boldsymbol{\Phi}_i - \boldsymbol{\Phi}_{i-1}}, \quad r_i^{HO} = \frac{\boldsymbol{\Phi}^{HO,L}_{i+\frac{1}{2}} - \boldsymbol{\Phi}_i}{\boldsymbol{\Phi}_i - \boldsymbol{\Phi}_{i-1}} \tag{66}$$

where the high-order face reconstruction $\boldsymbol{\Phi}^{HO,L}_{i+\frac{1}{2}}$ is obtained from WENO5 interpolations (Jiang and Shu, 1996). It will be shown in the results section that this reconstruction strategy is not diffusive enough for liquid cases. Instead, a standard MUSCL reconstruction with minmod limiter is used

$$r_i^{TVD} = \max[0, \min(1, r_i)]. \tag{67}$$

As mentioned in Sections 4.2 and 4.3, the conservative variables $\mathbf{Y}$ are not directly interpolated at the face. As it is often done in the literature, they are reconstructed from the interpolated values of primitive variables. For the ten-equation model of this work, the interpolated variables required to reconstruct $\mathbf{Y}$ at the face are $(\alpha_p, \alpha_p^\star, u_f, u_p, T_f, T_a, T_p, p_f, \Theta_p, k_f)$.

The two-fluid model with added mass needs additional care in the reconstruction of the left and right states. More specifically, the reconstruction of volume fraction and density for both phases needs to be consistent to retrieve a non-disturbing pressure scheme. The reconstruction of volume fractions must respect

$$\alpha^\star_{f,i+\frac{1}{2}} = 1 - \alpha^\star_{p,i+\frac{1}{2}}. \tag{68}$$

Moreover, it is demonstrated in Appendix A that a consistent reconstruction for ρ_e is based on Eq. (2):

$$\rho^K_{e,i+\frac{1}{2}} = \frac{\alpha^K_{p,i+\frac{1}{2}}}{\alpha^{\star,K}_{p,i+\frac{1}{2}}}\rho_p + \left(1 - \frac{\alpha^K_{p,i+\frac{1}{2}}}{\alpha^{\star,K}_{p,i+\frac{1}{2}}}\right)\frac{p_{f,i+\frac{1}{2}} + \gamma_a p_{\infty,a}}{(\gamma_a - 1)e^K_{a,i+\frac{1}{2}}} \tag{69}$$

where superscript $K = L$ or R. The fluid pressure at the face $p_{f,i+\frac{1}{2}}$ corresponds only to the thermodynamic pressure such that Eq. (27) is true at each face:

$$p_{f,i+\frac{1}{2}} = \hat{p}_{f,i+\frac{1}{2}} - \frac{2}{3}\rho_{f,i+\frac{1}{2}} k_{f,i+\frac{1}{2}}, \tag{70}$$

with $\hat{p}_{f,i+\frac{1}{2}}$, $\rho_{f,i+\frac{1}{2}}$ and $k_{f,i+\frac{1}{2}}$ obtained from the HLLC scheme detailed in Section 4.3. This tight coupling between phases is paramount to the stability of the method.

4.2. AUSM$^+$-up scheme for the particle subsystem

The particle subsystem is written as:

$$\mathbf{Y}^{n+1}_{p,i} = \mathbf{Y}^{n}_{p,i} - \frac{\Delta t}{\Delta x}(\boldsymbol{f}_{p,i+\frac{1}{2}} - \boldsymbol{f}_{p,i-\frac{1}{2}}) \tag{71}$$

with

$$\mathbf{Y}_p = \begin{bmatrix} \rho_p\alpha_p \\ \rho_a\alpha_a \\ \rho_e\alpha_p^\star u_p \\ \rho_e\alpha_p^\star K_p \\ \rho_p\alpha_p e_p \\ \rho_a\alpha_a e_p \end{bmatrix}, \quad \boldsymbol{f}_p = \begin{bmatrix} \rho_p\alpha_p u_p \\ \rho_a\alpha_a u_p \\ \rho_e\alpha_p^\star(u_p^2 + \Theta_p + P_p) \\ \rho_e\alpha_p^\star(K_p + \Theta_p + P_p)u_p \\ \rho_p\alpha_p e_p u_p \\ \rho_a\alpha_a e_a u_p \end{bmatrix}. \tag{72}$$

The particle phase has its own equation of state where compressibility is a result of volume-fraction variations:

$$p_p = \rho_e \alpha_p^\star (\Theta_p + P_p) = EOS(\rho_e \alpha_p^\star, \alpha_p, \Theta_p). \tag{73}$$

The compaction wave speed a_p (analogous to the fluid speed of sound) can be computed from (Serna and Marquina, 2005)

$$a_p = \frac{1}{\rho_p} \sqrt{\rho_p \left.\frac{\partial p_p}{\partial \alpha_p}\right|_{\Theta_p} + \frac{2}{3}\frac{\Theta_p}{\alpha_p^2} \left.\frac{\partial p_p}{\partial \Theta_p}\right|_{\alpha_p}^2}. \tag{74}$$

However, Eq. (74) was developed for standard particle systems where pressure is defined such that $p_p = EOS(\alpha_p, \Theta_p)$ with constant ρ_p.

In our model, p_p also depends on fluid quantities, and a_p is computed from the eigenvalues of the particle subsystem to account for added mass. There are three eigenvalues equal to u_p and two eigenvalues that are not necessarily symmetric such that

$$a_p = \max(u_p - \lambda_{p,min}, \lambda_{p,max} - u_p). \tag{75}$$

In gas–particle systems where $\rho_e \approx \rho_p$, computing a_p from Eq. (74) or Eq. (75) is almost equivalent. However, ρ_e can become several orders of magnitude higher than ρ_p in bubbly flows and a_p needs to be computed from the eigenvalues of the system to avoid stability issues.

The main difficulty arises from the granular pressure p_p in Eq. (73), which can take values from zero (low volume fraction) to extremely large (near the packing limit when P_f prevails). In other words, the Riemann solver needs to deal with a pressure-less system and a low-Mach limit, which are both known to cause problems with standard Godunov-type solvers (Guillard and Nkonga, 2017). AUSM-family schemes (Liou and C.J. Steffen, 1993) are perfect candidates to handle the pressure-less limit as they split the flux into convective and pressure parts, and have the capability to solve near-vacuum flows (Liou, 1996). Moreover, extra diffusion terms are introduced in the AUSM$^+$up scheme to handle the low-Mach limit (Liou, 2006).

The AUSM$^+$-up scheme applied to the particle subsystem leads to the particle flux vector:

$$\boldsymbol{f}_p = \dot{\mathbf{m}}_p \boldsymbol{\Psi}_p + \boldsymbol{\Pi}_p \tag{76}$$

with the mass flux vector $\dot{\mathbf{m}}_p = (\dot{m}_p, \dot{m}_a, \dot{m}_p^\star, \dot{m}_p^\star, \dot{m}_p, \dot{m}_a)$, the quantity vector $\boldsymbol{\Psi}_p = (1, 1, u_p, K_p, e_p, e_a)$, and the pressure flux $\boldsymbol{\Pi}_p = (0, 0, p_p, p_p u_p, 0, 0)$. In the original AUSM scheme (Liou, 2006), the pressure flux is only non-zero for momentum, while enthalpy was transported for energy. In Houim and Oran (2016), the pseudo-thermal energy Θ_p is solved, and the pressure term for energy is not included in the flux. Here, the kinetic energy K_p is solved and requires to consider a pressure flux consistent between momentum and energy; hence, the unconventional term $p_p u_p$ in $\boldsymbol{\Pi}_p$. Another modification of the AUSM$^+$-up scheme resides in the definition of mass fluxes as both particle and added mass are tracked and transported with the same speed u_p:

$$\dot{m}_p = \rho_p \alpha_p u_p, \quad \dot{m}_a = \rho_a \alpha_a u_p \tag{77}$$

The total mass flux used to transport velocity u_p and kinetic energy K_p is then defined with the discretely consistent expression $\dot{m}_p^\star = \dot{m}_p + \dot{m}_a$.

The quantity vector is upwinded using the face Mach number $\mathcal{M}_p^{1/2}$:

$$\boldsymbol{\Psi}_p = \begin{cases} \boldsymbol{\Psi}_p^L, & \text{if } \mathcal{M}_p^{1/2} \le 0 \\ \boldsymbol{\Psi}_p^R, & \text{otherwise.} \end{cases} \tag{78}$$

Following the choices of Houim and Oran (2016), the left and right reconstructions $\boldsymbol{\Psi}_p^L$ and $\boldsymbol{\Psi}_p^R$ of $\boldsymbol{\Psi}_p$ are obtained from the interpolation of primitive variables $(\alpha_p, \alpha_p^\star, u_p, \Theta_p, T_p)$ and the face value of $\rho_f^{1/2}$ from the HLLC solver detailed hereafter.

The mass flux $\dot{m}_p^\star$ is computed from $\mathcal{M}_p^{1/2}$ as

$$\dot{m}_p^\star = a_p^{1/2} \mathcal{M}_p^{1/2} \begin{cases} \rho_e^L \alpha_p^{\star,L}, & \text{if } \mathcal{M}_p^{1/2} \le 0 \\ \rho_e^R \alpha_p^{\star,R}, & \text{otherwise} \end{cases} \tag{79}$$

with $\rho_e^{L/R}$ obtained from Eq. (69). The conventional mass flux $\dot{m}_p$ can also be obtained for α_p:

$$\dot{m}_p = a_p^{1/2} \mathcal{M}_p^{1/2} \begin{cases} \rho_p \alpha_p^L, & \text{if } \mathcal{M}_p^{1/2} \le 0 \\ \rho_p \alpha_p^R, & \text{otherwise,} \end{cases} \tag{80}$$

which corresponds to $(\alpha_p \rho_p / \alpha_p^\star \rho_e) \dot{m}_p^\star$ computed from face reconstructions $\alpha_p^{L/R}$, $\alpha_p^{\star L/R}$, and $\rho_e^{L/R}$. This ensures that α_p and $\alpha_p^\star$ are transported at the same speed.

The face compaction wave speed $a_p^{1/2}$ is computed as in Houim and Oran (2016) from the following mass average:

$$a_p^{1/2} = \sqrt{\frac{\rho_e^L \alpha_p^{\star,L} (a_p^L)^2 + \rho_e^R \alpha_p^{\star,R} (a_p^R)^2}{\rho_e^L \alpha_p^{\star,L} + \rho_e^R \alpha_p^{\star,R}} + \epsilon} \tag{81}$$

with $\epsilon = 10^{-10}$ a small number to avoid division by zero when the compaction wave speed is null. In the same manner, the face density is defined as

$$\rho_e^{1/2} = \frac{\rho_e^L \alpha_p^{\star,L} + \rho_e^R \alpha_p^{\star,R}}{2}, \tag{82}$$

and the left, right, and averaged Mach numbers are computed from $a_p^{1/2}$:

$$\mathcal{M}_p^L = \frac{u_p^L}{a_p^{1/2}}, \qquad \mathcal{M}_p^R = \frac{u_p^R}{a_p^{1/2}}, \qquad \bar{\mathcal{M}}_p^2 = \frac{(u_p^L)^2 + (u_p^R)^2}{2(a_p^{1/2})^2}. \tag{83}$$

Finally, the face Mach number $\mathcal{M}_p^{1/2}$ related to the convective flux and the face pressure $\mathcal{P}_p^{1/2}$ related to the pressure flux are computed from the split Mach number and pressure polynomials $\mathscr{M}_4^{\pm}$ and $\mathscr{P}_5^{\pm}$ (Liou, 1996) as

$$\mathcal{M}_p^{1/2} = \mathscr{M}_4^+(\mathcal{M}_p^L) + \mathscr{M}_4^-(\mathcal{M}_p^R) - \frac{k_p}{f_a}\max(1-\sigma\bar{\mathcal{M}}_p^2, 0)\frac{p_p^R - p_p^L}{(\rho_e^{1/2}+\epsilon)(a_p^{1/2})^2}, \tag{84}$$

$$\mathcal{P}_p^{1/2} = \mathscr{P}_5^+(\mathcal{M}_p^L)p_p^L + \mathscr{P}_5^-(\mathcal{M}_p^R)p_p^R - 2k_u f_a \mathscr{P}_5^+(\mathcal{M}_p^L)\mathscr{P}_5^-(\mathcal{M}_p^R)\rho_e^{1/2}(a_p^{1/2}-\epsilon)(u_p^R - u_p^L) \tag{85}$$

with $k_p = 0.25$, $k_u = 0.75$, $f_a = 1$, and $\sigma = 1$ parameters controlling the pressure and velocity diffusion terms. The pressure flux for energy is then computed as $a_p^{1/2}\mathcal{M}_p^{1/2}\mathcal{P}_p^{1/2}$.

Some primitive variables $\mathbf{X}_p = (\alpha_p^\star, u_p, \Theta)$ are also required to compute non-conservative term and are upwinded with $\mathcal{M}_p^{1/2}$.

4.3. HLLC scheme for the fluid subsystem

The fluid subsystem is written as

$$\mathbf{Y}_{f,i}^{n+1} = \mathbf{Y}_{f,i}^n - \frac{\Delta t}{\Delta x}(\boldsymbol{f}_{f,i+\frac{1}{2}} - \boldsymbol{f}_{f,i-\frac{1}{2}}) \tag{86}$$

with

$$\mathbf{Y}_f = \begin{bmatrix} \rho_f\alpha_f^\star \\ \rho_f\alpha_f^\star u_f \\ \rho_f\alpha_f^\star E_f \\ \rho_f\alpha_f^\star k_f \end{bmatrix}, \qquad \boldsymbol{f}_f = \begin{bmatrix} \rho_f\alpha_f^\star u_f \\ \rho_f\alpha_f^\star u_f^2 + \rho_f\alpha_p^\star\left(\Theta_p\mathbf{I} + \mathbf{R}\right) \\ \alpha_f^\star(\rho_f E_f + \hat{p}_f)u_f + \rho_f\alpha_p^\star\mathbf{r} \\ \rho_f\alpha_f^\star k_f u_f \end{bmatrix}. \tag{87}$$

To solve this system, the particle-phase volume fraction is frozen during a time step and the fluid-phase volume fraction is reconstructed at the face from Eq. (68) to ensure consistency between the amount of particles seen from both phases.

While the flux contains the fluid pressure, it does not correspond to a standard Euler step weighted by the volume fraction $\alpha_f^\star$. For this reason, the numerical scheme must be able to compute the reconstruction of fluid pressure apart from the flux, which is not the case for the standard HLL scheme. Hence, the HLLC scheme is used to reconstruct the variables $\mathbf{X}_f = (\rho_f, u_f, \hat{p}_f, E_f, k_f)$.

The HLLC scheme is a three-wave approximate Riemann solver with four intermediate states such that

$$\mathbf{X}_f^{HLLC} = \begin{cases} \mathbf{X}_f^L, & \text{if } 0 \le S^L \\ \mathbf{X}_f^{L\star}, & \text{if } S^L \le 0 \le S^\star \\ \mathbf{X}_f^{R\star}, & \text{if } S^\star \le 0 \le S^R \\ \mathbf{X}_f^R, & \text{if } S^R \le 0 \end{cases} \tag{88}$$

with $\mathbf{X}_f^K$ left ($K = L$) and right ($K = R$) states reconstructed at the face, and $\mathbf{X}_f^{K\star}$ the intermediate states at the left and right sides of the contact wave computed as described below. The left and right states of $\mathbf{X}_f$ are obtained from interpolation of the primitive variables (u_f, T_f, p_f, k_f) and the EOS given by Eq. (26). In the standard HLLC method, the left and right wave speed S^K are reconstructed as in HLL (Einfeldt, 1988) or from the Roe-average (Einfeldt et al., 1991). In our solver, they are retrieved from the minimum and maximum eigenvalues of the full system in Eq. (64), which accounts for added mass, and the contact wave speed $S^\star$ is computed as in Batten et al. (1997):

$$S^L = \lambda_{f,min}, \quad S^R = \lambda_{f,max}, \quad S^\star = \frac{\hat{p}_f^R - \hat{p}_f^L + \rho_f^L u_f^L(S^L - u_f^L) - \rho_f^R u_f^R(S^R - u_f^R)}{\rho_f^L(S^L - u_f^L) - \rho_f^R(S^R - u_f^R)}. \tag{89}$$

The contact pressure can then be computed as

$$\hat{p}_f^\star = \frac{1}{2}[\hat{p}_f^L + \hat{p}_f^R + \rho_f^L(S^L - u_f^L)(S^\star - u_f^L) + \rho_f^R(S^R - u_f^R)(S^\star - u_f^R)]. \tag{90}$$

Finally, the left and right states apart from the contact wave can be computed as $\mathbf{X}_f^{K\star} = (\rho_f^{K\star}, S^\star, \hat{p}_f^\star, E_f^{K*}, k_f^K$, such that

$$\rho_f^{K\star} = \rho_f^K\frac{S^K - u_f^K}{S^K - S^\star}, \quad E_f^{K\star} = E_f^K + \frac{S^\star\hat{p}_f^\star - S^K\hat{p}_f^K}{\rho_f^K(S^K - u_f^K)}. \tag{91}$$

The scheme results in an upwinding of P_{pfp}. This is a necessary but not sufficient component of the method for stability of the model as P_{pfp} is also discontinuous across a particle front. While this term is not significant in dense gas–particle systems, it needs to be treated carefully for bubbly flows.

The fluid fluxes are finally assembled using $\alpha^\star_{f,i+\frac{1}{2}} = 1 - \alpha^\star_{p,i+\frac{1}{2}}$, $\Theta_{p,i+\frac{1}{2}}$, $u_{p,i+\frac{1}{2}}$ from the AUSM$^+$-up scheme and $\mathbf{X}_{f,i+\frac{1}{2}}$ from Eq. (88) such that

$$\boldsymbol{f}_{f,i+\frac{1}{2}} = \begin{pmatrix} \alpha^\star_{f,i+\frac{1}{2}}\rho_{f,i+\frac{1}{2}}u_{f,i+\frac{1}{2}} \\ \alpha^\star_{f,i+\frac{1}{2}}\rho_{f,i+\frac{1}{2}}u_{f,i+\frac{1}{2}}u_{f,i+\frac{1}{2}} + \hat{p}_{f,i+\frac{1}{2}} \\ \alpha^\star_{f,i+\frac{1}{2}}\rho_{f,i+\frac{1}{2}}E_{f,i+\frac{1}{2}}u_{f,i+\frac{1}{2}} + \alpha^\star_{f,i+\frac{1}{2}}u_{f,i+\frac{1}{2}}\hat{p}_{f,i+\frac{1}{2}} \\ \alpha^\star_{f,i+\frac{1}{2}}\rho_{f,i+\frac{1}{2}}k_{f,i+\frac{1}{2}}u_{f,i+\frac{1}{2}} \end{pmatrix}, \tag{92}$$

4.4. Non-conservative terms

Finally the non-conservative terms are reconstructed. It has been shown in the literature that the discretization of those terms need to be defined carefully to be compatible with the flux reconstruction. In practice, the face reconstructions defined in Sections 4.2 and 4.3 are used to compute $\boldsymbol{h}(\mathbf{Y})$. The treatment of F_{pf} is slightly different as upwinding of u_p and u_f can be in different directions due to the decoupling between particle and fluid systems. To avoid any issue with mixed reconstructions, the direction of upwinding of u_f used to compute u_{pf} and u_v is the same as $\alpha_p^\star$. In the our numerical experiments, this choice tended to stabilize the simulations. This issue could be treated differently by building an approximate Reimann solver for the full system of equations with a more complex wave pattern as presented in Furfaro and Saurel (2015) for a more conventional seven-equation system.

4.5. Final algorithm and time integration

All the terms described in the previous sections can be assembled to constitute the hyperbolic step $\mathcal{H}_{\Delta t}$ with the final discretization given in Appendix A. Here is a summary of the algorithm to compute fluxes:

1. Compute the fluid variable vector $\mathbf{X}_{f,i+\frac{1}{2}}$ and the thermodynamic pressure $p_{f,i+\frac{1}{2}}$ from Eqs. (26) and (88) following the HLLC procedure.
2. Compute the particle fluxes $\boldsymbol{f}_{p,i+\frac{1}{2}}$ from Eq. (76), the particle volume fraction $\alpha^\star_{p,i+\frac{1}{2}}$, the particle velocity $u_{p,i+\frac{1}{2}}$ and the granular temperature $\Theta_{p,i+\frac{1}{2}}$ following the AUSM$^+$-up procedure.
3. Assemble the fluid fluxes $\boldsymbol{f}_{f,i+\frac{1}{2}}$ from Eq. (92) the non-conservative terms using the quantities computed in the previous steps.

In order to remain stable with WENO5 reconstructions, a RK3-SSP scheme is employed that also preserves realizability (Vié et al., 2015) and the time integration proceeds as follows (Spiteri and Ruuth, 2002):

$$\begin{aligned}
\mathbf{Y}^{(1)} &= \mathbf{Y}^n - \Delta t\, \partial_x \boldsymbol{f}(\mathbf{Y}^n) + \Delta t\, \boldsymbol{h}(\mathbf{Y}^n),\\
\mathbf{Y}^{(2)} &= \frac{3}{4}\mathbf{Y}^n + \frac{1}{4}[\mathbf{Y}^{(1)} - \Delta t\, \partial_x \boldsymbol{f}(\mathbf{Y}^{(1)}) + \Delta t\, \boldsymbol{h}(\mathbf{Y}^{(1)})],\\
\mathbf{Y}^{n+1} &= \frac{1}{3}\mathbf{Y}^n + \frac{2}{3}[\mathbf{Y}^{(2)} - \Delta t\, \partial_x \boldsymbol{f}(\mathbf{Y}^{(2)}) + \Delta t\, \boldsymbol{h}(\mathbf{Y}^{(2)})].
\end{aligned} \tag{93}$$

The overall time step is based on the maximum particle and fluid eigenvalues. The eigenvalues are more complicated than for the standard Baer–Nunziato equations and are found numerically using a subroutine for the Jacobian matrix that was generated symbolically. Hyperbolicity of a similar system has been demonstrated in Fox et al. (2020), which is applicable for the system presented in this paper. Indeed, the transport of internal energy of particles introduced in this work does not modify the eigenvalues of the nine-equation model as it is simply advected by $\mathbf{u}_p$. More details on the eigenvalues of this system can be found in Fox (2025). Note that Θ_p can become negative in regions of vanishing α_p near shocks where u_{pf} becomes very high. In this case, the value of Θ_p is set back to a small positive value 10^{-12} and the total kinetic energy of particles K_p is updated to absorb this change of Θ_p without impacting the mean kinetic energy $u_p^2/2$.

5. Additional physics

For the additional physical terms related to local mass, momentum and energy transfers between scales and phases, source terms $\boldsymbol{s}(\mathbf{Y})$ are treated using a time splitting of the different relaxation mechanisms sorted by timescales. This results in solving a succession of linear ODEs that can be solved analytically. A first advantage of this approach is to impose strict conservation of quantities during the integration time step. Also, it alleviates the time constraint, which can be very restrictive in the case of dramatic shocks interacting with a dispersed phase. For each term, a separate semi-analytical solution is derived and used to update the conserved variables in each phase. The semi-analytical solutions of drag $\mathcal{S}^D$, added mass $\mathcal{S}^{AM}$, heat transfer $\mathcal{S}^{HT}$ and particle collision–friction $\mathcal{S}^{CF}$ are detailed in Appendix B.

The local source term operator is then written as:

$$\mathcal{S}_{\Delta t} = \mathcal{S}^{CF}_{\Delta t}\mathcal{S}^{D}_{\Delta t}\mathcal{S}^{HT}_{\Delta t}\mathcal{S}^{AM}_{\Delta t} \tag{94}$$

with Δt the time step from the hyperbolic step.

6. Numerical results and examples

This section consists of various 1-D test cases to assess the solver's capabilities to handle a large range of Mach numbers, volume fractions, and material-density ratios between the fluid and particles. For this purpose, the fluid can be either air or water with properties given in Table 2. The particle phase can be solid particles or non-deformable spherical bubbles with properties provided in the corresponding test case. In the solver, the minimum value of the particle-phase volume fraction α_ϵ is set at a small value (10^{-12}) to avoid division by zero. Therefore, the quantities for the disperse phase are only displayed when $\alpha_p > 10^{-8}$ to provide meaningful values. In all test cases, if not specified, the model parameters take the values $e_c = 1$, $\Delta_f = 0.01$, $p_{fr} = 533,333$ kg/m/s^2, $\alpha_{max} = 0.65$ and the CFL number takes the value of 0.5. The added mass is always initialized to the equilibrium value $\alpha_a^\star$ defined in Section 2.1.

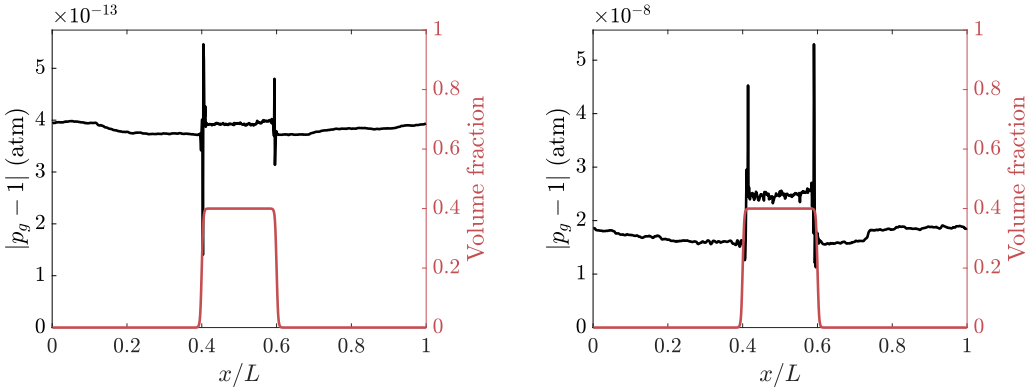
In general, for gas–particle test cases, the model formulation, numerical solver, and results are very similar to those reported in Houim and Oran (2016). In contrast, the bubbly flow cases are more challenging for the hyperbolic solver because of the strong phase coupling due to buoyancy and added mass. Unlike for gas–particle flows wherein the largest eigenvalues can arise in the particle phase, this does not occur for bubbly flows. However, it has been shown that particle–fluid–particle terms ensure hyperbolicity for any relevant physical parameters in Fox (2025).

6.1. Assessment of the hyperbolic solver

In this section, we verify that the proposed solver satisfies a few basic properties of the continuum mode.

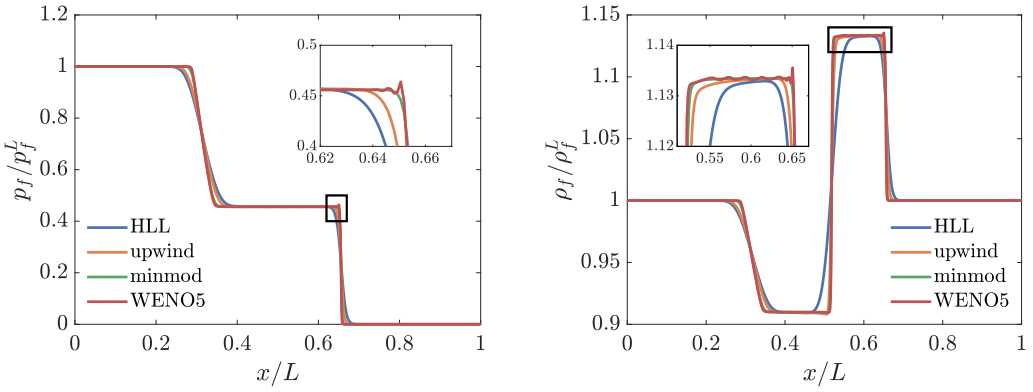
Table 2
Default physical properties and limiter scheme of the fluids used for test cases (added-mass properties are taken equal to fluid properties).

Fluid	γ_f	$p_{\infty,f}$ (GPa)	ν_f (Pa s)	$C_{p,f}$ (J/kg/K)	λ_f (m^2/s)	limiter
Air	1.4	0	$2.12 \cdot 10^{-5}$	1005	$2.56 \cdot 10^{-2}$	WENO5 (Eq.(66))
Water	4.4	0.6	10^{-6}	5250	$6 \cdot 10^{-2}$	minmod (Eq.(67))



(a) Air (b) Water

Fig. 2. Gas-phase pressure (black) and particle volume fraction (red) for the mechanical equilibrium after one box length travel with $N_x = 800$. Figure *(a)* shows results for air and figure *(b)* for water.



(a) Fluid pressure (b) Fluid density

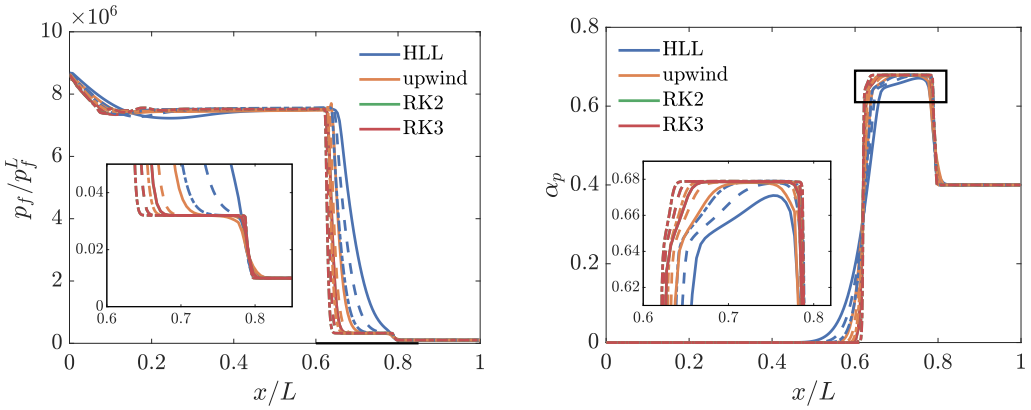
Fig. 3. Fluid pressure *(a)* and fluid density *(b)* of a water shock at $t = 79$ µs for the different flux reconstructions.

6.1.1. *Mechanical equilibrium of a moving particle dust*

First, the pressure non-disturbing condition is demonstrated on the advection of a dust cloud in two different media: *(i)* air with equal velocities $u_p = u_f = 100$ m/s; *(ii)* water with equal velocities $u_p = u_f = 1000$ m/s. For mechanical equilibrium to be retrieved, equal temperatures $T_p = T_f = 300$ K and fluid pressure $p_f = 1$ atm are taken at initialization while Θ_p and k_f are set to zero. As velocities are equal and granular temperature is zero, the pfp terms do not act on the flow. The dust cloud of width 0.2 m and $\alpha_p = 0.4$ is centered in a periodic domain with length $L = 1$ m. In Fig. 2, the pressure errors and the volume fractions are displayed after one box length travel with a grid refinement of $N_x = 800$. The pressure errors are of order 10^{-13}, which demonstrates the consistency of the hyperbolic system discretization. Yet, it is interesting to notice that the pressure error increases for a liquid fluid. This is due to the stiffened gas EoS amplifying round-off errors.

6.1.2. *High-order reconstructions for liquid flows*

While the high-order methodology to compute primitives from the limiter Eq. (66) with WENO5 reconstruction is stable for perfect gas ($p_\infty = 0$). It was found to cause oscillations near shocks for liquids in the numerical experiments. This is due to the combination of the HLLC scheme and a very low-dissipative reconstruction that produces small oscillations, which are amplified due to the high wave speeds in a stiffened gas flow with large p_∞. To show this behavior, a shock tube with pure water ($\alpha_p = 0$) is now considered such that the system results in the well-known Euler equations for a single-species fluid. The aim is to show that the issue arises even if non-conservative terms and source terms are absent. The domain length is $L = 1$ m and the left and right states are separated by a diaphragm located at $x = 0.5$ m. The left and right states are taken as $p_f^L = 10^9$ Pa, $p_f^R = 10^5$ Pa, $\rho_f^L = \rho_f^R = 1000$ kg/m^3. Different slope limiters are compared for the current methodology: low-order upwind from $r_i^{TVD} = 0$ (and forward Euler for temporal integration), minmod from Eq. (67) and WENO5 from Eq. (66). Also, the methodology of Fox et al. (2020) is added as a reference (HLL).



(a) Gas pressure (b) Volume fraction

Fig. 4. Gas-phase pressure (*a*) and particle volume fraction (*b*) for the dense shock tube case at $t = 100$ µs using HLL (blue), upwind, high-order with RK2 (green covered by the red curve) and high-order with RK3 (green). The three levels of grid refinement are $N_x = 200$ (——), 400 (- - -) and 800 (-···-).

In Fig. 3, it is clear that even without particles, the liquid shock shows oscillations near the discontinuity when using WENO5 face reconstructions. If MUSCL reconstructions are used instead, oscillations are canceled. For reference while the stiffness of the discontinuity is correctly captured compared to the reference scheme HLL and the first-order upwind reconstruction. Therefore, the minmod limiter is used instead of the limited WENO5 reconstruction to remove this dispersive behavior. This methodology will be used in the results implying water as a fluid.

6.1.3. Mesh convergence on a two-phase shock tube

The computational scheme is next compared to the first-order HLL scheme employed in Fox et al. (2020) for a two-phase shock tube implying air as a fluid. The first-order upwind scheme introduced in the previous section is also studied to compare the features of the proposed solver setting aside the order of reconstruction of the quantities. Moreover, a RK2-SSP time integration scheme is tested with high-order reconstructions and compared to the RK3-SSP scheme presented in Section 4.5 with WENO5 face reconstructions. The verification test case is taken from Houim and Oran (2016) where a strong shock interacts with a dense particle cloud. It consists of a shock tube where the left chamber contains pure high-pressure air, while the right chamber is filled with a mixture of air and particles. Hence, the case allows to test the capability of the solver to capture a shock, a contact discontinuity and a particle front in the presence of strong dynamics.

The particle concentration is chosen to be high, such that the volume-fraction front approaches the packing limit as the shock pushes the particle phase to the right of the domain. The method should be able to remain stable when P_{fr} controls the eigenvalues. Moreover, the strong shock creates a sudden heating of the gas that is partially absorbed by the particle phase, which has the effect of greatly weakening the shock. The computational domain length is $L = 0.06$ m, and the left and right states are separated by a diaphragm located at $x = 0.03$ m. Solid particles are defined with $\rho_p = 1470$ kg/m^3, $d_p = 5$ µm, and $C_{p,p} = 987$ J/kg/K. The left and right states for the gas are $p_f^L = 100$ atm, $p_f^R = 1$ atm, $T_{f,p}^L = T_{f,p}^R = 300$ K, $\alpha_f^L = 1$, and $\alpha_f^R = 0.6$. The three schemes are compared for three levels of grid refinement: $N_x = 200$, 400, and 800.

The fluid pressure and particle volume fraction at $t = 100$ µs are shown in Fig. 4 for HLL, upwind, high-order with RK2 and RK3. It is clear from both the pressure and volume fraction that the new scheme combining AUSM$^+$-up and HLLC is better at capturing discontinuities. Even though first-order reconstructions are used, the new scheme is able to capture the particle front accurately. More quantitatively, the HLL scheme requires $N_x = 800$ to give the same degree of detail compared to our proposed algorithm and upwind reconstruction with $N_x = 200$ on this test case. The effect of using high-order reconstructions is also visible for this test case, as shown in the previous section. Finally, RK2 and RK3 schemes are giving exactly the same accuracy. This is not so surprising in 1D shock-tube problems, where the spatial errors are concentrated in the discontinuity vicinity, where slope limiters reduce the spatial accuracy to remain TVD. Differences would be more visible in multidimensional test cases where complex fluid structure can develop and require an accurate temporal resolution to properly capture the convective effects in the hyperbolic operator. Overall, all schemes converge with mesh refinement, and the high-order schemes capture discontinuities even with a coarse grid.

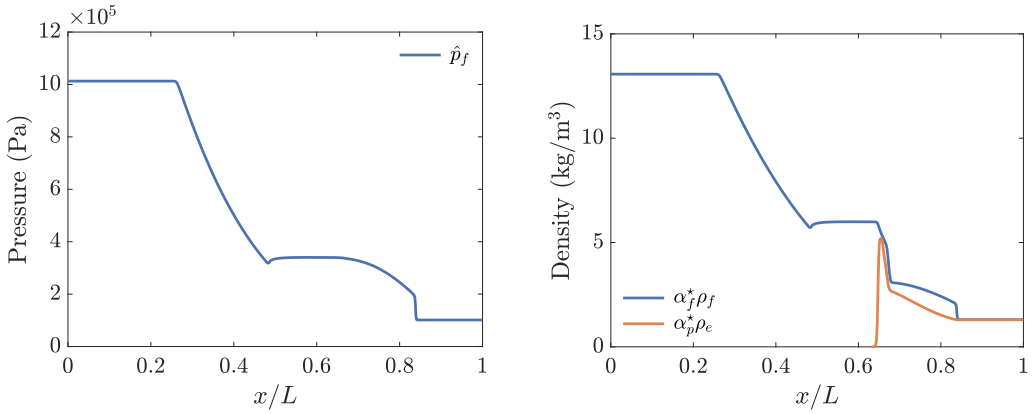
Now that the proposed solver has been verified and mesh convergence has been assessed on a demanding test case, the remaining numerical examples are computed using the high-order scheme with RK3 at the highest mesh refinement, $N_x = 800$ if not specified.

6.2. Air shock wave interacting with a particle cloud

In this example, a dilute shock tube and the dense shock tube presented above are investigated. In general, gas–particle flows are simpler to simulate than bubbly flows due to the relatively small effect of buoyancy. However, due to differences in the model formulation (e.g., the frictional source and pfp pressure), results from the two-fluid model in Table 1 can be expected to exhibit some differences with the model in Houim and Oran (2016) for gas–particle flows.

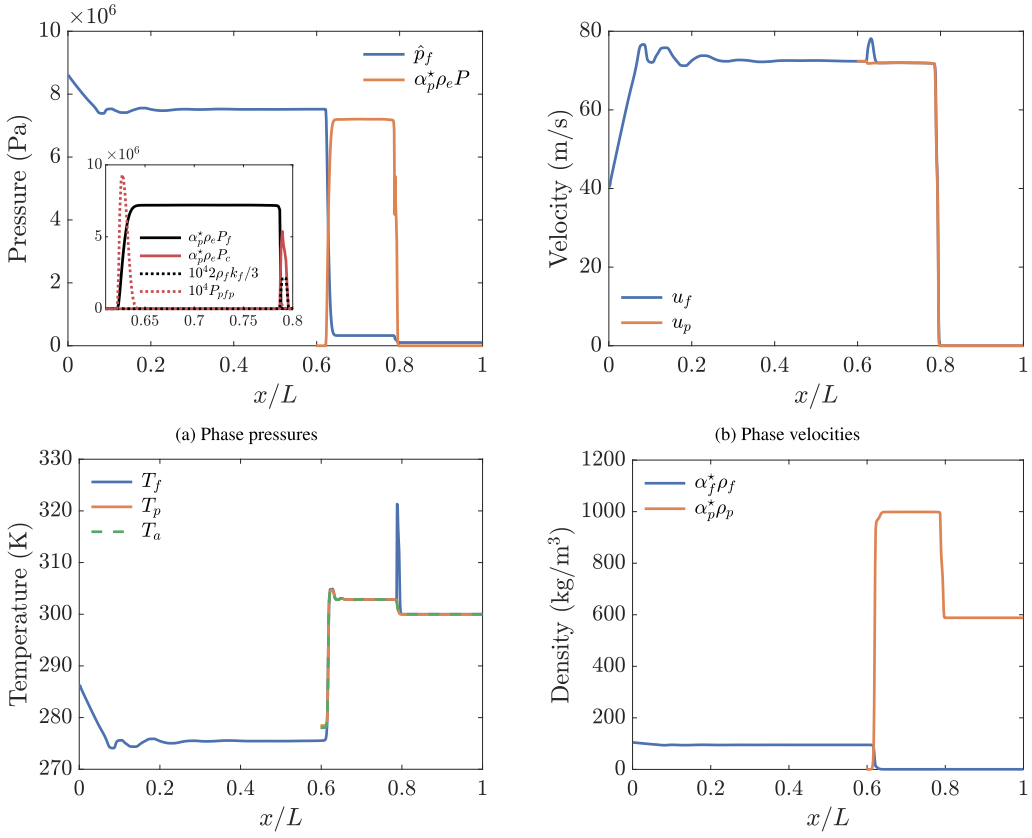
6.2.1. Dilute shock tube

For this example, the right state contains a very dilute particle phase such that the solver must handle an almost pressure-less system. The domain length is $L = 0.257798$ m and the left and right states are separated by a diaphragm located at $x = 0.129$ m to reproduce the same setup as in Houim and Oran (2016). Even though no exact solution exists for this case, results can be compared to the results presented in Houim and Oran (2016). Solid particles are defined with $\rho_p = 2500$ kg/m^3, $d_p = 10$ µm, and $C_{p,p} = 720$ J/kg/K. The left and right states are $p_f^L = 10$ atm, $p_f^R = 1$ atm, $T_{f,p}^L = T_{f,p}^R = 270$ K, $\alpha_p^L = 0$, and $\alpha_p^R = 5.172 \cdot 10^{-4}$. The quantities of interest are plotted at $t = 184$ µs in Fig. 5.



(a) Gas pressure (b) Phase densities

Fig. 5. Gas-phase pressure (a) and phase densities (b) for the dilute shock tube at $t = 184$ µs.



(a) Phase pressures (b) Phase velocities

(c) Phase temperature (d) Phase densities

Fig. 6. Phase pressures (a), velocities (b), temperatures (c), and densities (d) for the dense shock tube at $t = 1$ ms.

As the shock progresses in the gas–particle mixture, it experiences momentum and heat loss because of the drag force and heat transfer. As noted in Houim and Oran (2016), the hook-like structure of $\hat{p}_f$ near the initial shock position is well captured. The main difference with Houim and Oran (2016) is the profile of particle mass density near $x = 0.15$, which is higher with our model. This is likely due to the unsteady force F_{pf} that accelerates the particles near the shock and contact surface. Overall, the physical behavior is retrieved and showcases the performance of the proposed scheme near pressure-less conditions.

6.2.2. *Dense shock tube*

The problem introduced in Section 6.1 is repeated with more details to better describe the weakening of the shock, the balance of particle pressure in the dense part of the granular phase, and the effect of drag on phase velocities. In Fig. 6, the phase pressures, velocities, temperatures and mass densities are provided. The solutions agree well with Houim and Oran (2016), apart from the granular pressure, which experiences an abrupt decrease at the particle front.

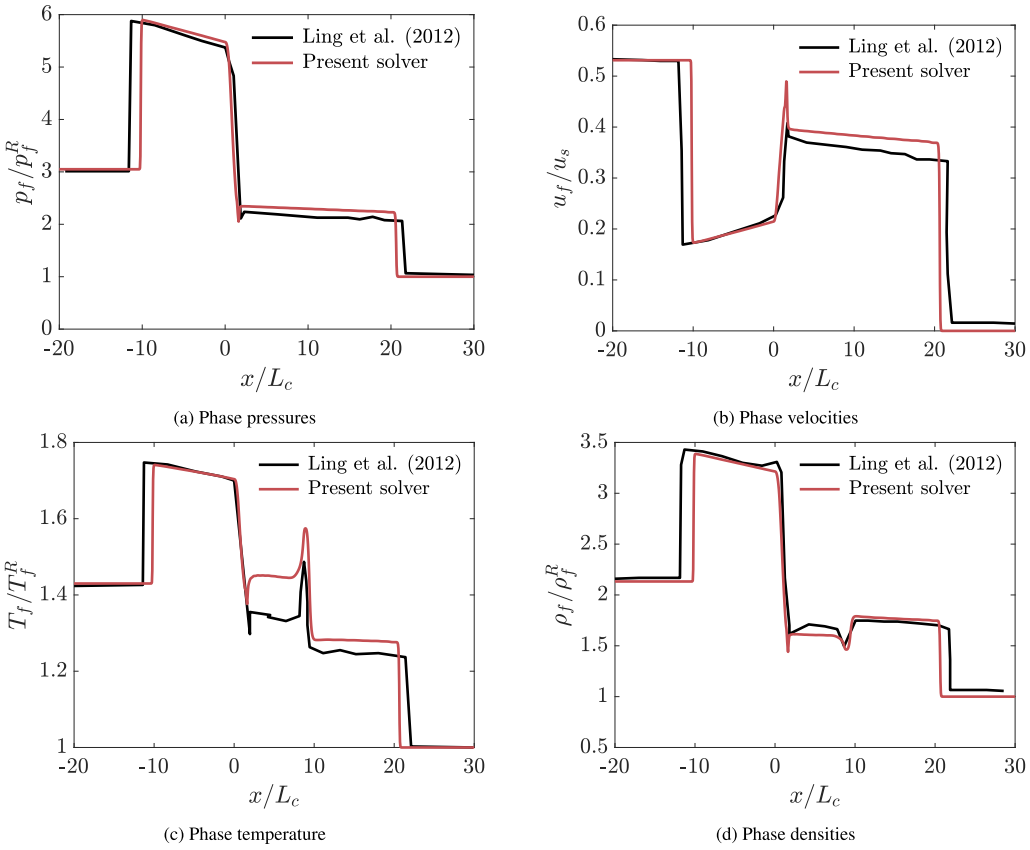


Fig. 7. Profiles of fluid pressure (a), velocity (b), temperature (c), and density (d) at $t = 24L_c/u_s$. Results of the present solver are compared with the Euler–Lagrange simulations of Ling et al. (2012).

The collisional and frictional parts of the particle-phase pressure are shown separately in the inset figure of Fig. 6(a), which are clearly separated at $\alpha_p = \alpha_{max}$. The frictional pressure is controlling where the volume fraction approaches the packing limit, but decreases abruptly at the particle front. This behavior is due to the different functional forms used for the frictional pressure in the present model compared to what was used in Houim and Oran (2016). By increasing the parameter Δ_f, the transition zone is larger and frictional pressure is applied at lower volume fractions. In the same inset figure, the pressure term due to PTKE also rises because of particle agitation Θ_p transferred into k_f. The P_{pfp} tensor is also pictured and only acts at high relative velocity at the particle front where it tries to push particles backward. However, the effect of $2\rho_f k_f/3$ and P_{pfp} are negligible for a gas-particles system because $\rho_p/\rho_f \gg 1$. This is why they are rescaled by 10^4 such that they are comparable to the particle-phase pressure in the inset figure. In Fig. 6(c), the temperature profiles showcase the heating of particles through the shock, which explains the weakening of the shock with an amount equivalent to what was observed in Houim and Oran (2016).

Finally, it is interesting to notice that the present model is stable and without particle-pressure oscillations, even when $e_c = 1$. In Houim and Oran (2016), strong oscillations were observed and additional dissipation was needed in the numerical scheme to control them. In the current model, the granular temperature Θ_p is strongly damped to negligible values in the frictional regime by the frictional source term included in H_p, a term that is not present in Houim and Oran (2016). Moreover, we use a sharper cut-off between the frictional and collisional regimes in the definition of P_{fr} (see individual particle pressure contribution in Fig. 6(a)), and compute the particle-phase eigenvalues directly from the Jacobian matrix. As a result, the proposed hyperbolic solver is robust with $e_c = 1$ without additional dissipation even when $\alpha_p > \alpha_{max}$.

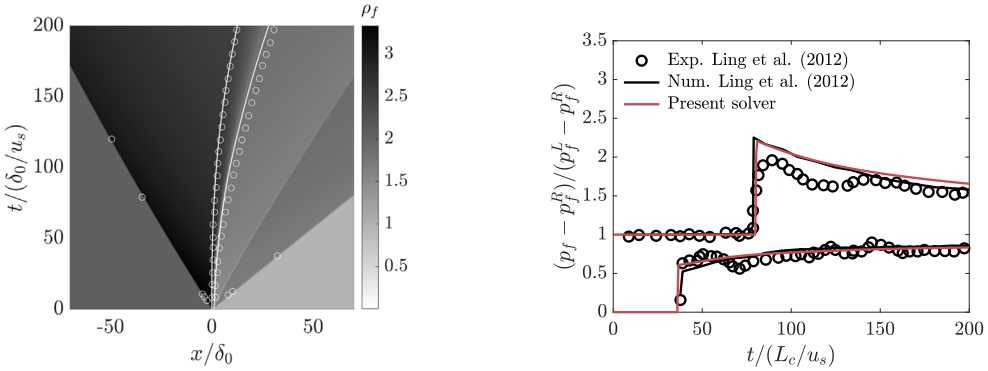
6.3. Shock-induced dispersal of particle curtains

Next, the gas–particle solver is validated against experimental data. The most commonly used experiment is the interaction of a shock with a particle curtain which has been used for decades to validate numerical solvers against the experiment of Rogue et al. (1998). Here, a combination of experimental and Euler–Lagrange simulations of Ling et al. (2012) is chosen.

The computational domain is $[-100L_c, 100L_c]$ with the curtain width $L_c = 2$ mm and the particle concentration $\alpha_p = 0.21$. Non-reflective boundary conditions are used to match the experiments and the mesh is refined enough with $N_x = 4000$ resulting in a curtain resolution of 20 cells. The shock is produced by a fluid pressure ratio with the diaphragm located upstream of the curtain at $x_d = -25L_c$. The fluid pressure p_f^L and temperature T_f^L of the post-shock (left) state are deduced from the shock Mach number $\mathcal{M}_s = 1.66$ and ambient conditions in the pre-shock region $T_f^R = 297$ K and $p_f^R = 84.1$ kPa.

First, the instant obtained from the solver at $t = 24L_c/us$ is compared with the Euler–Lagrange simulations of Ling et al. (2012). In Fig. 7, the overall behavior is well reproduced by the solver with a difference of temperature inside the particle curtain. This might be due to the difference of modeling of the heat transfer which is less detailed in the Euler–Lagrange solver with no heating through particle collisions.

For the curtain dynamics, x–t diagram of density with isocontours of α_p from the simulation are represented in Fig. 8(a) and experimental results of Ling et al. (2012) are added with scattered circles. The time is defined such that $t = 0$ represents the moment where the shock hits the upstream



(a) x–t diagram of density with isocontours of α_p.

(b) Evolution of fluid pressure in time at two probes.

Fig. 8. Shock-induced dispersal of particle curtains illustrated with the x–t diagram of density *(a)* and the comparison of the spreading rate with experimental data *(b)*. Isocontour of α_p is displayed in white for the present solver and experimental data of Ling et al. (2012) are represented with scattered circles representing the curtain edges and the reflected/transmitted shocks.

Table 3
Two configurations from Behrendt et al. (2022) tested for a water shock in a fixed particle array. τ corresponds to the transit time d_p/u_s.

Configuration	$\mathcal{M}_s$	α_p	p_f^L (MPa)	ρ_f^L (kg/m^3)	u_f^L (m/s)	τ (ns)
LM10	1.22	0.1	477.73	1136.1	241.2	50.39
HM20	3	0.2	7823.7	1487.8	1606.5	20.49

edge of the curtain. Then a part of the shock wave is transmitted, while a reflected shock wave travels back upstream. It can be seen that the reflected shock, the transmitted shock and the upstream edge of the curtain are well reproduced while the downstream edge is transported more slowly than observed in the experiments. The same trend was observed when comparing Euler–Lagrange simulations and experimental data (Ling et al., 2012), which was related to shock–boundary-layer interactions. This difference could also be explained by the difficulty to detect the curtain edges with accuracy in both numerical results and experimental images at the downstream position where the particles are highly spread (Houim and Oran, 2016). Moreover, the particle phase is not likely to be monodisperse, and polydispersity can also plays a significant role in the dispersion of the curtain Fox et al. (2024). The displacement of the downstream edge is mainly controlled by the drag force, which is coarsely modeled in this work. A detailed study of the effect of the drag model on the physical behavior of the particles can be found in a related work using a previous iteration of this multifluid model (Boniou and Fox, 2023).

6.4. Underwater shock wave in a fixed particle array

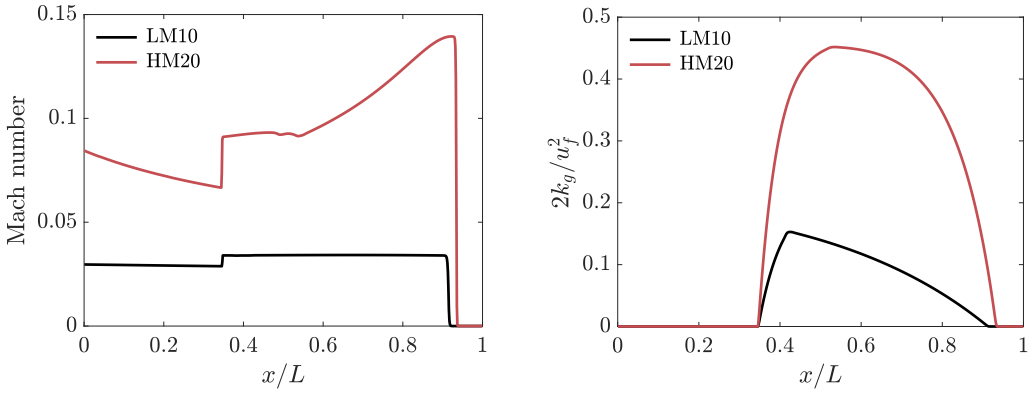
Now that gas–particle systems have been investigated, the interaction of an underwater shock with fixed particles is considered. The main particularity of a water shock compared to a gas shock is that the resulting post-shock Mach number is subsonic. The setup is based on the PRDNS study presented in Behrendt et al. (2022) where a random array of fixed insulated particles is defined in a part of the domain and a strong shock is initialized upstream of the array. In the model, the absence of motion and heat transfer of particles is obtained by fixing $\rho_p = 10^9$ kg/m^3 such that timescales for exchanges between particles and liquid is very large compared to the timescale of the flow. As drag timescale will be very large, the added mass does not have time to come back to equilibrium volume fraction $\alpha_a^\star$. However, added mass will still interact with the liquid phase through heat transfer H_{af} and pressure equilibrium $p_g = p_a$ in the shock. The mean particle diameter is $d_p = 100$ µm and heat transfer is deactivated as in the PRDNS. The domain is taken as $L = 26d_p$, while the particle array is located at $x_p = 9d_p$ and extends until the right end of the domain. The shock is initially positioned one diameter before the particle array $x_s = 8d_p$.

In Behrendt et al. (2022), two Mach numbers and two volume fractions are simulated, leading to four configurations. Here, the two configurations, referred as LM10 and HM20, are tested and compared to the PRDNS results. The volume fraction and post-shock (left) state corresponding to the two configurations are summarized in Table 3. The pre-shock (right) state is defined as $p_f^R = 1$ atm, $\rho_f^R = 998.0$ kg/m^3. The simulations stop at $t = 16\tau$ with $\tau = d_p/u_s$ the transit time. In the simulations, large fluctuations of the fluid velocity (corresponding to PTKE) can be observed for HM20, while only small interactions with the particle array are observed for LM10. To compare with the PRDNS results, the Mach number u_f/a_f and dimensionless PTKE $2k_f/u_f^2$ are shown in Fig. 9.

The results in Fig. 9(a) compare well with the Mach contour provided in Behrendt et al. (2022) where reflections are only observed for HM20. The velocity fluctuations, corresponding to PTKE in the model, are well reproduced in the 1-D simulation wherein the dimensionless PTKE reaches 50% for HM20 while it only reaches 15% for LM10 in Fig. 9(b).

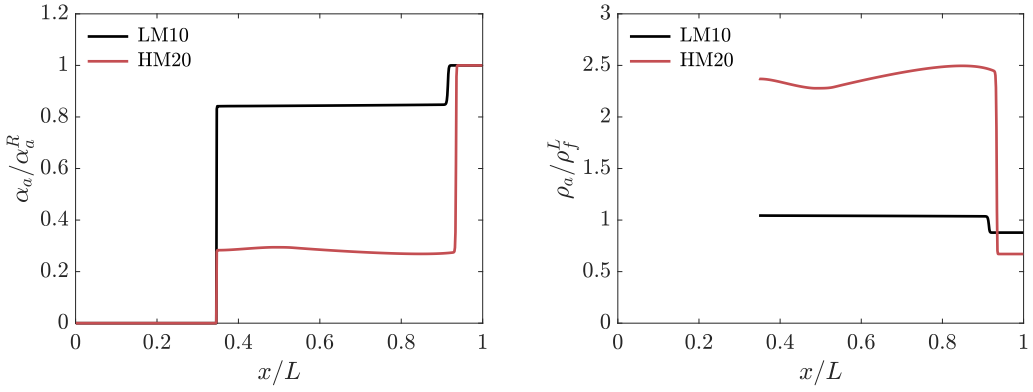
Some information about added mass is also provided in Fig. 10 to complete the study. In Fig. 10(a), the volume fractions of added mass α_a increases in the particle array as a result of fluid-density variations observed in Fig. 10(b). This illustrates the effects of added mass even for a case where the particle locations are fixed. The predicted magnitude of this effect can be modified using the parameter C_a in the definition of added-mass relaxation timescale τ_a.

Finally, as velocity fluctuations are very high in this test case, it is interesting to study more deeply the role of PTKE transport in the ten-equation system. This is done by comparing the pressure of the fluid when PTKE transport is switched off to the previous results. From the definition of pressure Eq. (27), PTKE increases the sound speed for the same thermodynamic properties. However, it is clear from Fig. 11 that for HM20 the shock is less strong when PTKE is transported and the pressure (even with the additional isotropic term) is not as large as for the case without



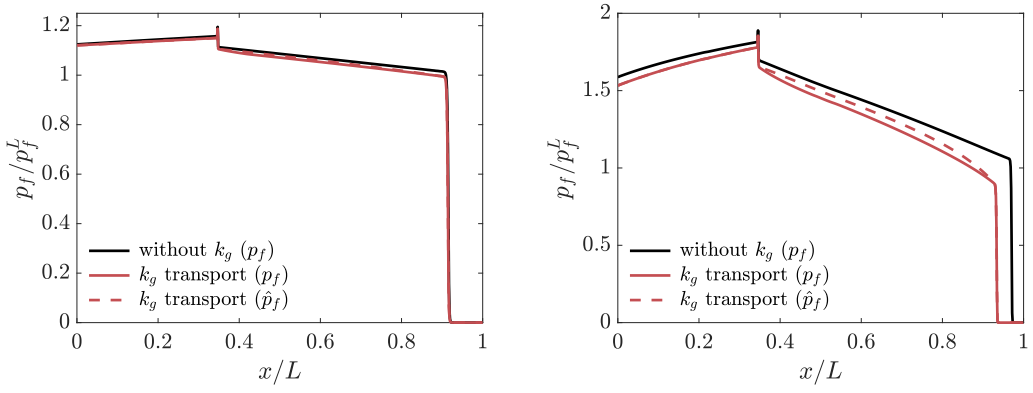
(a) Fluid Mach number (b) PTKE

Fig. 9. Fluid Mach number (*a*), PTKE compensated by fluid kinetic energy (*b*) for the two configurations LM10 and HM20 at $t = 16\tau$.



(a) Volume fraction (b) Density

Fig. 10. Volume fraction (*a*) and density (*c*) of the added-mass phase for the two configurations LM10 and HM20 at $t = 16\tau$.



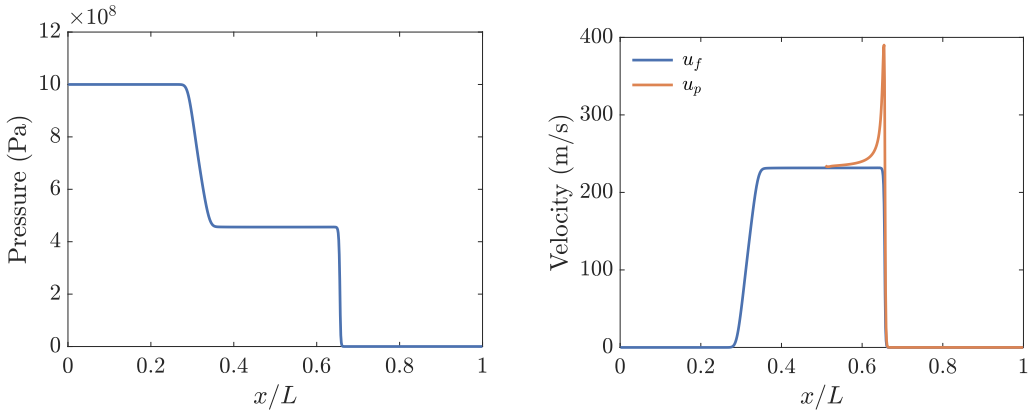
(a) LM10 (b) HM20

Fig. 11. Effect of PTKE modeling on the pressure for two configurations LM10 (*a*) and HM20 (*b*) at $t = 16\tau$.

PTKE. Indeed, PTKE is an additional contribution for the pressure but it also stores a part of the drag work instead of converting all of it into heat. As a result, the fluid heating is not as high as it would be without transfer of energy into PTKE and as a result, the shock is not enhanced as much. While, the weakening of a shock was due to the heat transfer from fluid to particles in the gas-solid system studied in Section 6.2, the weakening – or absence of enhancement – of the shock is now due to the transfer of a part of the drag work into PTKE instead of fluid heating. This highlights the richness of this ten-equation system, which contains both mechanisms in its modeling.

6.5. Shock wave in a bubbly flow

The final numerical example is adapted from the shock tube test cases of Section 6.2, but the fluid is now water and the particle phase has a low material density such that it mimics non-deformable spherical bubbles. The domain length is $L = 1$ m and the left and right states are separated by a



(a) Fluid pressures (b) Phase velocities

Fig. 12. Fluid pressure (*a*) and phase velocities (*b*) for low aeration at $t = 79$ µs.

diaphragm located at $x = 0.5$ m to reproduce the same setup as in Furfaro and Saurel (2015). Particles are defined with $\rho_p = 1$ kg/m^3, $d_p = 500$ µm, and $C_{p,p} = 987$ J/kg/K. The left and right states are taken as $p_f^L = 10^9$ Pa, $p_f^R = 10^5$ Pa, $\rho_f^L = \rho_f^R = 1000$ kg/m^3. The left state is pure water, while the right state corresponds to a bubbly flow with different aeration levels α_p^R. Note that a proper bubble model would use the internal bubble pressure to determine the bubble diameter and material density. Nevertheless, the setup is still relevant to show the capability of the solver to handle large material-density ratios between the fluid and particles, and the resultant slip velocity. In such systems, the buoyancy terms become important and hyperbolicity is only preserved thanks to the pfp pressure (Fox et al., 2020).

6.5.1. Low aeration

As done for the gas–particle system, a first test is defined with a low volume fraction of $\alpha_p^R = 10^{-4}$. The pressure and phase velocities at $t = 79$ µs are displayed in Fig. 12. As expected, the bubble swarm is too dilute to impact the water shock. The shock wave then crosses the bubble swarm without decelerating. From the velocity profiles, it is interesting to notice that the slip velocity is especially large compared to the gas–particle systems. This is due to the predominant buoyancy terms in bubbly flows. At the shock front, the pressure gradient is very large and creates a strong acceleration of the bubble swarm. The bubble velocity subsequently relaxes back to the fluid velocity behind the shock due to drag. Similar, but not as strong, behavior is observed for smaller bubbles. As expected, the predicted values of Θ_p are small in bubbly flows due to the small Stokes number.

6.5.2. High aeration

In this case, the same setup is used with a volume fraction of $\alpha_p^R = 0.4$. The fluid pressure, phase velocities, temperatures and mass densities at $t = 79$ µs are displayed in Fig. 13. In contrast to the gas–particle dense case, the shock is not weakened by the dense bubble swarm. On the contrary, it is accelerated because of the reduction of cross section experienced by water as α_p increases. However, the same velocity peak observed in gas–particle flows appears at the upstream edge of bubble swarm. The pressure gradient is smoothed at the shock front, which does not produce an acceleration comparable to the dilute case and the pure case studied in Section 6.1.2. As a result, the bubble volume fraction does not increase significantly as was observed for the heavy particles such that the volume fraction never reach the friction regime. The inset figure in Fig. 13(a) shows the collisional part of granular pressure, which has a smoother rise than for the gas-particle system as a consequence of the smoothing of the shock front. The pressure term due to PTKE is also rising with the same trend due to the transfer between k_f and Θ_p. For bubbles, $\rho_p/\rho_f \ll 1$ such that the magnitude of P_{pfp} is higher than the stresses due to velocity fluctuations. Finally, the temperatures of the bubbles and added mass in Fig. 13(c) do not reach the temperature of the fluid because the timescale related to heat transfer is too large compared to the acoustic timescale.

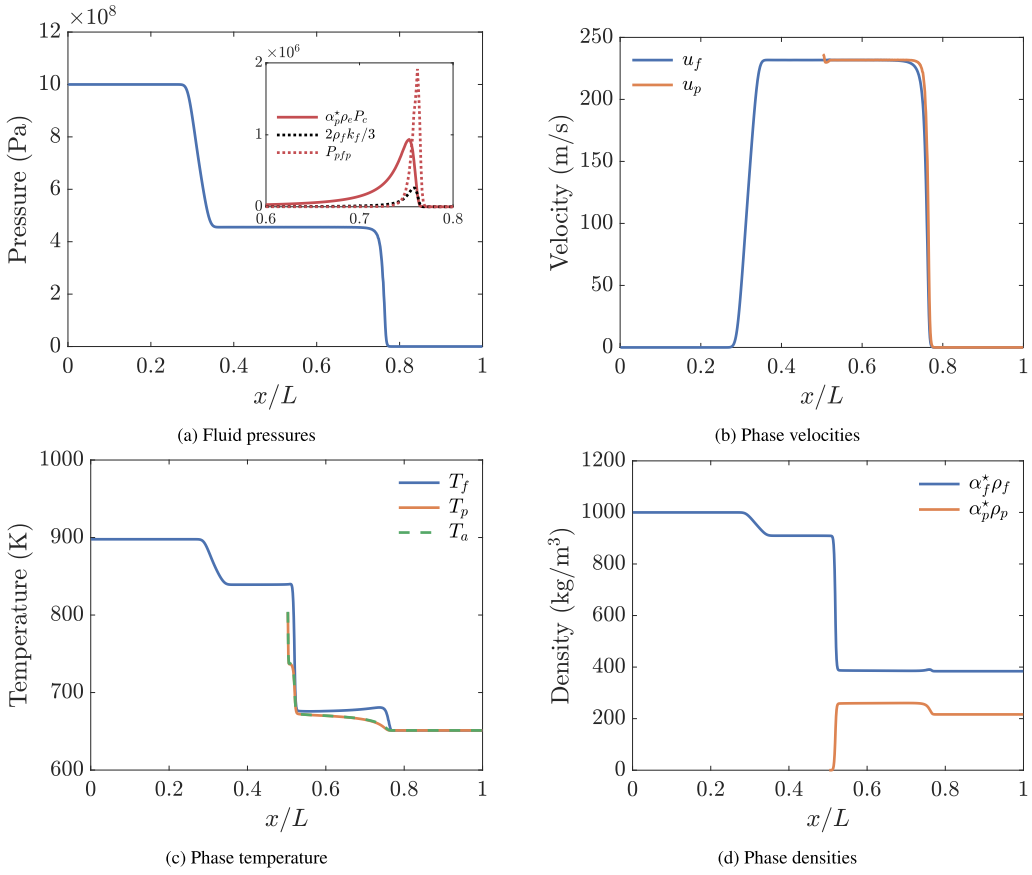
This last test case demonstrates the capability of the solver to simulate water–bubble systems where added mass and buoyancy terms are predominant. In gravity-driven systems, large volume-fraction regions (e.g., foams) can be produced, leading to a frictional-pressure contributions. However, such flow dynamics occur on much longer time scales than the shock waves investigated here.

7. Conclusions

In this work, a hyperbolic macroscale model for high-speed, fully compressible, monodisperse fluid–particle flows with added mass and fluid-phase pseudoturbulence has been developed and tested. The particle-phase equations have been derived from a kinetic model with terms for the collisional, frictional, and pfp-pressure contributions to the particle pressure. The kinetic model accounts as well for the particle-phase and added-mass-phase internal energy to capture heat transfer with the fluid phase.

For the numerical solver, the resulting ten-equation macroscale model was discretized with a higher-order method along the lines of the pioneering work of Houim and Oran (2016). The main issue was to adapt the combination of AUSM$^+$-up and HLLC initially proposed in the context of a seven-equation, two-fluid model without added mass to the present model. This has been done by treating carefully the additional non-conservative terms with a consistent discretization. Moreover, the complex coupling of particle- and fluid-phase quantities through the added mass was treated by combining consistently the fluid and particle reconstructions and computing the eigenvalues directly from the Jacobian of the spatial fluxes. While this is not necessary for gas-solid systems, this treatment is crucial for systems involving liquids.

The first numerical study assessed the treatment of the non-conservative terms on the advection of a dust cloud in air and water where the mechanical equilibrium is retrieved discretely. Then, it has been demonstrated that for a stiffened-gas EoS applied to liquids, a minmod limiter was mandatory to cancel spurious oscillations near a strong shock. Finally, the mesh convergence of the solver has been verified on a challenging



(a) Fluid pressures

(b) Phase velocities

(c) Phase temperature

(d) Phase densities

Fig. 13. Fluid pressure (*a*), phase velocities (*b*), temperatures (*c*), and mass densities (*d*) for high aeration at $t = 79$ µs.

gas–particle test case, including a granular phase reaching the packing limit and a strong shock in the fluid phase. The advantages of the proposed method have been showcased in comparison to the first-order HLL scheme used in Fox et al. (2020) and the impact of the time integration has been discussed.

Then, the numerical examples presented in the results section demonstrate the capability of the numerical solver to handle a wide range of two-phase systems of supersonic flows, dilute-to-dense granular phases, and gas–particle to bubbly flows. Generally speaking, the macroscale model is able to reproduce the global behavior of particle-resolved DNS, Euler–Lagrange simulations and experiments from the literature.

In future work, the kinetic-based model and the numerical scheme developed in this work could be extended to handle reacting flows, chemical species in the fluid and particle phases can be modeled as done in Houim and Kuo (2011), Houim and Oran (2016). This could be combined with the phase-change solver proposed in Boniou et al. (2024) and the polydisperse-solver in Fox et al. (2024) for a full description of the dispersed phase. On another note, the main advantage of the proposed framework is to handle arbitrary density ratios which made the study of compressible bubbly flows possible. However, to extend the application range to other bubbly flows, it would be useful to develop a low-Mach-number solver for the macroscale model that neglects the acoustic waves in the fluid phase (Passalacqua and Fox, 2011). For such flows, it will be necessary to include constitutive relations for the viscous and thermal fluxes (Chauchat and Médale, 2014; Chauchat, 2018; Berzi et al., 2020; Jenkins and Larcher, 2023), which have no effect on the hyperbolicity of the macroscale model (Fox et al., 2020). These additions would allow the macroscale model to be used for gravity-driven and/or pressure-driven multiphase flows such as bubble columns, sedimentation, and fluidized beds.

CRediT authorship contribution statement

Victor Boniou: Writing – original draft, Methodology, Investigation. **Rodney O. Fox:** Writing – original draft, Methodology, Funding acquisition. **Frédérique Laurent:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

When preparing this work, ROF was supported by the 2022–2023 Fulbright–Tocqueville Distinguished Chair Award. The Government of the United States or any agency representing it has not endorsed the conclusions or approved the contents of this publication.

Appendix A. Discrete mechanical equilibrium

This appendix is devoted to the numerical treatment of the hyperbolic system and the consistent reconstruction on cell faces. First, let us write the discrete hyperbolic system for the full model:

$$(\partial_t \alpha_p \rho_p)_i = -\frac{\dot{m}_{p,i+\frac{1}{2}} - \dot{m}_{p,i-\frac{1}{2}}}{\Delta x} \tag{A.1}$$

$$(\partial_t \alpha_a \rho_a)_i = -\frac{\dot{m}_{a,i+\frac{1}{2}} - \dot{m}_{a,i-\frac{1}{2}}}{\Delta x} \tag{A.2}$$

$$(\partial_t \alpha^\star_f \rho_f)_i = -\frac{\alpha^\star_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} u_{f,i+\frac{1}{2}} - \alpha^\star_{f,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} u_{f,i-\frac{1}{2}}}{\Delta x} \tag{A.3}$$

$$\begin{aligned}(\partial_t \alpha^\star_p \rho_e u_p)_i = &-\frac{\dot{m}^\star_{p,i+\frac{1}{2}} u_{p,i+\frac{1}{2}} + p_{p,i+\frac{1}{2}} - \dot{m}^\star_{p,i-\frac{1}{2}} u_{p,i-\frac{1}{2}} - p_{p,i-\frac{1}{2}}}{\Delta x} \\ &- \alpha^\star_{p,i} \frac{\hat{p}_{f,i+\frac{1}{2}} - \hat{p}_{f,i-\frac{1}{2}}}{\Delta x} - F_{pf,i}\end{aligned} \tag{A.4}$$

$$\begin{aligned}(\partial_t \alpha^\star_f \rho_f u_f)_i = &-\frac{\alpha^\star_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} u_{f,i+\frac{1}{2}} u_{f,i+\frac{1}{2}} - \alpha^\star_{f,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} u_{f,i-\frac{1}{2}} u_{f,i-\frac{1}{2}}}{\Delta x} \\ &- \frac{\alpha^\star_{p,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} (\Theta_{p,i+\frac{1}{2}} + R_{i+\frac{1}{2}}) - \alpha^\star_{p,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} (\Theta_{p,i-\frac{1}{2}} + R_{i-\frac{1}{2}})}{\Delta x} \\ &- \alpha^\star_{f,i} \frac{\hat{p}_{f,i+\frac{1}{2}} - \hat{p}_{f,i-\frac{1}{2}}}{\Delta x} + F_{pf,i}\end{aligned} \tag{A.5}$$

$$\begin{aligned}(\partial_t \alpha^\star_p \rho_e K_p)_i = &-\frac{\dot{m}^\star_{p,i+\frac{1}{2}} K_{p,i+\frac{1}{2}} + p_{p,i+\frac{1}{2}} u_{p,i+\frac{1}{2}} - \dot{m}^\star_{p,i-\frac{1}{2}} K_{p,i-\frac{1}{2}} - p_{p,i-\frac{1}{2}} u_{p,i-\frac{1}{2}}}{\Delta x} \\ &- \alpha^\star_{p,i} u_{p,i} \frac{\hat{p}_{f,i+\frac{1}{2}} - \hat{p}_{f,i-\frac{1}{2}}}{\Delta x} - D_{pf,i}\end{aligned} \tag{A.6}$$

$$(\partial_t \alpha_p \rho_p e_p)_i = -\frac{\dot{m}_{p,i+\frac{1}{2}} e_{p,i+\frac{1}{2}} - \dot{m}_{p,i-\frac{1}{2}} e_{p,i-\frac{1}{2}}}{\Delta x} \tag{A.7}$$

$$(\partial_t \alpha_a \rho_a e_a)_i = -\frac{\dot{m}_{a,i+\frac{1}{2}} e_{a,i+\frac{1}{2}} - \dot{m}_{a,i-\frac{1}{2}} e_{a,i-\frac{1}{2}}}{\Delta x} \tag{A.8}$$

$$\begin{aligned}(\partial_t \alpha^\star_f \rho_f E_f)_i = &-\frac{\alpha^\star_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} E_{f,i+\frac{1}{2}} u_{f,i+\frac{1}{2}} - \alpha^\star_{f,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} E_{f,i-\frac{1}{2}} u_{f,i-\frac{1}{2}}}{\Delta x} \\ &- \frac{\alpha^\star_{p,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} r_{i+\frac{1}{2}} - \alpha^\star_{p,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} r_{i-\frac{1}{2}}}{\Delta x} \\ &- \hat{p}_{f,i} \frac{\alpha^\star_{p,i+\frac{1}{2}} u_{p,i+\frac{1}{2}} - \alpha^\star_{p,i-\frac{1}{2}} u_{p,i-\frac{1}{2}}}{\Delta x} - D_{pf,i}\end{aligned} \tag{A.9}$$

$$(\partial_t \alpha^\star_f \rho_f k_f)_i = -\frac{\alpha^\star_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} k_{f,i+\frac{1}{2}} u_{f,i+\frac{1}{2}} - \alpha^\star_{f,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} k_{f,i-\frac{1}{2}} u_{f,i-\frac{1}{2}}}{\Delta x} - \frac{2}{3} \alpha^\star_{f,i} \rho_{f,i} k_{f,i} \frac{u_{f,i+\frac{1}{2}} - u_{f,i-\frac{1}{2}}}{\Delta x} \tag{A.10}$$

Moreover, global mass conservation leads to the following identity:

$$\partial_t(\alpha^\star_p \rho_e + \alpha^\star_f \rho_f) = \partial_t(\alpha_p \rho_p + \alpha_f \rho_f). \tag{A.11}$$

From a discrete perspective, condition Eq. (A.11) can only be obtained by properly interpolating quantities at the faces.

By adding Eq. (A.2) and (A.3), and using the definition of $\dot{m}^\star_p$ from Eq. (79), we find

$$(\partial_t \alpha^\star_p \rho_e + \partial_t \alpha^\star_f \rho_f)_i = -\frac{\alpha^\star_{p,i+\frac{1}{2}} \rho_{e,i+\frac{1}{2}} u_{p,i+\frac{1}{2}} + \alpha^\star_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} u_{f,i+\frac{1}{2}} - \alpha^\star_{p,i-\frac{1}{2}} \rho_{e,i-\frac{1}{2}} u_{p,i-\frac{1}{2}} - \alpha^\star_{f,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} u_{f,i-\frac{1}{2}}}{\Delta x}. \tag{A.12}$$

The sum of particle and fluid continuity equations without added-mass (i.e., with $\alpha_a = 0$) gives

$$(\partial_t \alpha_p \rho_p + \partial_t \alpha_f \rho_f)_i = -\frac{\alpha_{p,i+\frac{1}{2}} \rho_p u_{p,i+\frac{1}{2}} + \alpha_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} u_{f,i+\frac{1}{2}} - \alpha_{p,i-\frac{1}{2}} \rho_p u_{p,i-\frac{1}{2}} - \alpha_{f,i-\frac{1}{2}} \rho_{f,i-\frac{1}{2}} u_{f,i-\frac{1}{2}}}{\Delta x}. \tag{A.13}$$

At mechanical equilibrium, $u_f = u_p = u_m$ and Eq. (A.11) is true discretely if the fluxes are equal at each face:

$$\alpha^\star_{p,i+\frac{1}{2}} \rho_{e,i+\frac{1}{2}} u_m + \alpha^\star_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} u_m = \alpha_{p,i+\frac{1}{2}} \rho_p u_m + \alpha_{f,i+\frac{1}{2}} \rho_{f,i+\frac{1}{2}} u_m, \tag{A.14}$$

which leads to a condition on $\alpha^\star_{p,i+\frac{1}{2}} \rho_{e,i+\frac{1}{2}}$:

$$\alpha^\star_{p,i+\frac{1}{2}} \rho_{e,i+\frac{1}{2}} = \alpha_{p,i+\frac{1}{2}} \rho_p + (\alpha_{f,i+\frac{1}{2}} - \alpha^\star_{f,i+\frac{1}{2}}) \rho_{f,i+\frac{1}{2}}. \tag{A.15}$$

If the volume-fraction reconstructions are consistent between the particle and fluid phases, then $\alpha^\star_{f,i+\frac{1}{2}} = 1 - \alpha^\star_{p,i+\frac{1}{2}}$ and the following condition can be written for $\rho_{e,i+\frac{1}{2}}$:

$$\rho_{e,i+\frac{1}{2}} = \frac{\alpha_{p,i+\frac{1}{2}}}{\alpha^\star_{p,i+\frac{1}{2}}}(\rho_p - \rho_{f,i+\frac{1}{2}}) + \rho_{f,i+\frac{1}{2}}. \tag{A.16}$$

Note that this condition is consistent with the definition of ρ_e in Eq. (2).

Appendix B. Analytic integration of source terms

In the numerical method presented in the main text, time splitting is used to solve for the different local source terms. As shown below, this splitting allows to reduce the complex equation system to simple ODEs with analytic solutions. Hence, the potential stability issues due to stiff source terms are avoided. The details on added-mass integration are found in the appendix of Boniou et al. (2024).

B.1. Drag source terms

A semi-analytical solution is used to update the drag terms in the 1-D momentum and energy balances, which amounts to solving the following system of ODEs:

$$\begin{aligned}
\partial_t \rho_f \alpha^\star_f &= 0 \\
\partial_t \rho_e \alpha^\star_p &= 0 \\
\partial_t \rho_f \alpha^\star_f u_f &= \frac{\rho_e \alpha^\star_p}{\tau_p}(u_p - u_f) \\
\partial_t \rho_e \alpha^\star_p u_p &= \frac{\rho_e \alpha^\star_p}{\tau_p}(u_f - u_p) \\
\partial_t \rho_f \alpha^\star_f k_f &= \frac{\rho_e \alpha^\star_p}{\tau_p}[3a\Theta_p - 2(1-a)k_f + u^2_{pf} - C_f k_f] \\
\partial_t \rho_e \alpha^\star_p K_p &= -\frac{\rho_e \alpha^\star_p}{\tau_p}[3a\Theta_p - 2(1-a)k_f + u_{pf}u_p]
\end{aligned}
\quad \Longrightarrow \quad
\begin{aligned}
r &= \frac{\rho_e \alpha^\star_p}{\rho_f \alpha^\star_f} \\
\partial_t(u_f - u_p) &= -\frac{1+r}{\tau_p}(u_f - u_p) \\
\partial_t(u_f + r u_p) &= 0 \\
\partial_t k_f &= \frac{r}{\tau_p}[3a\Theta_p - 2(1-a)k_f] + \frac{r}{\tau_p}[u^2_{pf}(t) - C_f k_f] \\
\partial_t \Theta_p &= \frac{1}{\tau_p}\left[\frac{4}{3}(1-a)k_f - 2a\Theta_p\right]
\end{aligned} \tag{B.1}$$

with r constant. The particle-phase internal energy e_p does not change due to drag, and the change in fluid-phase total energy $\rho_f \alpha^\star_f E_f$ is determined from conservation of total energy, i.e., $\rho_f \alpha^\star_f E_f + \rho_e \alpha^\star_p E_p$ is constant during drag. Note that the right-hand side for k_f has exchange terms and production/dissipation terms. Each part is handled separately.

Holding τ_p constant over the time step, the updated phase velocities are approximated by

$$\begin{aligned}
u_f(t) - u_p(t) &= (u_f(0) - u_p(0))e^{-(1+r)t/\tau_p} \\
u_f(t) + r u_p(t) &= u_f(0) + r u_p(0)
\end{aligned}
\quad \Longrightarrow \quad
\begin{aligned}
u_f(t) &= \frac{u_f(0)}{1+r}(1 + re^{-(1+r)t/\tau_p}) + \frac{r u_p(0)}{1+r}(1 - e^{-(1+r)t/\tau_p}) \\
u_p(t) &= \frac{u_f(0)}{1+r}(1 - e^{-(1+r)t/\tau_p}) + \frac{u_p(0)}{1+r}(r + e^{-(1+r)t/\tau_p})
\end{aligned} \tag{B.2}$$

and thus the source term for k_f is given by

$$u^2_{pf}(t) = u^2_{pf}(0)e^{-2(1+r)t/\tau_p}. \tag{B.3}$$

The evolution of $k_f(t)$ is written in two steps. First due to production and dissipation, it is written

$$\hat{k}_f = k_f(0)e^{-C_f rt/\tau_p} + \frac{r u^2_{pf}(0)}{2 + (2 - C_f)r}\left(e^{-C_f rt/\tau_p} - e^{-2(1+r)t/\tau_p}\right). \tag{B.4}$$

Second, the exchange terms form a linear system:

$$\partial_t \begin{bmatrix} k_f \\ \Theta_p \end{bmatrix} = -\frac{1}{\tau_p}\mathbf{A}\begin{bmatrix} k_f \\ \Theta_p \end{bmatrix} \qquad \text{where} \qquad \mathbf{A} = \begin{bmatrix} 2(1-a)r & -3ar \\ -\frac{4}{3}(1-a) & 2a \end{bmatrix}. \tag{B.5}$$

Since $0 < a_{min} \le a \le 1$, the two eigenvalues of $\mathbf{A}$ are non-negative: $\lambda_1 = 0$ and $\lambda_2 = 2r(1-a) + 2a$. Hence, the solution is

$$\begin{aligned}
k_f(t) &= \frac{1}{\lambda_2}\left[2r(1-a)\hat{k}_f e^{-\lambda_2 t/\tau_p} + 2a\hat{k}_f + 3ra\Theta_p(0)(1 - e^{-\lambda_2 t/\tau_p})\right], \\
\Theta_p(t) &= \frac{1}{\lambda_2}\left[2r(1-a)\Theta_p(0) + 2a\Theta_p(0)e^{-\lambda_2 t/\tau_p} + \frac{4}{3}(1-a)\hat{k}_f(1 - e^{-\lambda_2 t/\tau_p})\right].
\end{aligned} \tag{B.6}$$

Note that $2k_f(t) + 3r\Theta_p(t) = 2\hat{k}_f + 3r\Theta_p(0)$ is constant for the exchange terms, and the parameter a fixes the steady-state value of k_f.

B.2. Heat-transfer source term

A semi-analytical solution is used to update the heat transfer in the 1-D energy balances, which amounts to solving the following system of ODEs:

$$\begin{aligned}
\partial_t \rho_f \alpha_f^\star E_f &= 6\frac{\alpha_p^\star \lambda_f Nu_a}{d_a^2}(T_a - T_f) \\
\partial_t \rho_a \alpha_a e_a &= -6\frac{\alpha_p^\star \lambda_f Nu_a}{d_a^2}(T_a - T_f) \\
&\quad +6\frac{\alpha_p \lambda_a Nu_p}{d_a^2}(T_p - T_a) \\
\partial_t \rho_p \alpha_p e_p &= -6\frac{\alpha_p \lambda_a Nu_p}{d_p^2}(T_p - T_a)
\end{aligned} \quad \Longrightarrow \quad \begin{aligned}
\partial_t T_f &= 6\frac{\alpha_p^\star \lambda_f Nu_a \gamma_f}{\rho_f \alpha_f^\star C_{p,f} d_a^2}(T_a - T_f) \\
\partial_t T_a &= -6\frac{\alpha_p^\star \lambda_f Nu_a \gamma_a}{\rho_a \alpha_a C_{p,a} d_a^2}(T_a - T_f) \\
&\quad +6\frac{\alpha_p \lambda_a Nu_p \gamma_a}{\rho_a \alpha_a C_{p,a} d_p^2}(T_p - T_a) \\
\partial_t T_p &= -6\frac{\alpha_p \lambda_a Nu_p}{\rho_p \alpha_p C_{p,p} d_p^2}(T_p - T_a)
\end{aligned} . \tag{B.7}$$

The kinetic energy does not change during heat transfer, hence only internal energy of the fluid and the particles are modified. Conservation of total energy also states that $\rho_f \alpha_f^\star E_f + \rho_e \alpha_p^\star (K_p + e_p + e_a)$ is constant during heat transfer.

Holding all quantities but temperature constant over the time step, the updated phase temperatures are approximated by solving the following exponential matrix:

$$\mathbf{T}(t) = e^{A(t)}\mathbf{T}(0), \tag{B.8}$$

with vector and matrix defined as

$$\mathbf{T} = \begin{bmatrix} T_f(t) \\ T_a(t) \\ T_p(t) \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} -6\frac{\alpha_p^\star \lambda_f Nu_a \gamma_f}{\rho_f \alpha_f^\star C_{p,f} d_a^2} & 6\frac{\alpha_p^\star \lambda_f Nu_a \gamma_f}{\rho_f \alpha_f^\star C_{p,f} d_a^2} & 0 \\ 6\frac{\alpha_p^\star \lambda_f Nu_a \gamma_a}{\rho_a \alpha_a C_{p,a} d_a^2} & -6\frac{\alpha_p^\star \lambda_f Nu_a \gamma_a}{\rho_a \alpha_a C_{p,a} d_a^2} - 6\frac{\alpha_p \lambda_a Nu_p \gamma_a}{\rho_a \alpha_a C_{p,a} d_p^2} & 6\frac{\alpha_p \lambda_a Nu_p \gamma_a}{\rho_a \alpha_a C_{p,a} d_p^2} \\ 0 & 6\frac{\alpha_p \lambda_a Nu_p}{\rho_p \alpha_p C_{p,p} d_p^2} & -6\frac{\alpha_p \lambda_a Nu_p}{\rho_p \alpha_p C_{p,p} d_p^2} \end{bmatrix}. \tag{B.9}$$

The updated conservative variables $\rho_f \alpha_f^\star E_f$, $\rho_a \alpha_a e_a$ and $\rho_p \alpha_p e_p$ are then retrieved from the updated temperatures with $\rho_f \alpha_f^\star$, $\rho_a \alpha_a$, $\rho_p \alpha_p$, u_f, u_p and k_f constant:

$$\rho_f \alpha_f^\star E_f(t) = \rho_f \alpha_f^\star \left(\frac{1}{2}u_f^2 + \frac{C_{p,f}}{\gamma_f} T_f(t) + k_f \right), \tag{B.10}$$

$$\rho_a \alpha_a e_a(t) = \rho_a \alpha_a \frac{C_{p,a}}{\gamma_a} T_a(t), \tag{B.11}$$

$$\rho_p \alpha_p e_p(t) = \rho_p \alpha_p C_{p,p} T_p(t). \tag{B.12}$$

B.3. Collisional and frictional source terms

A semi-analytical solution is used to update the collisional source terms in the 1-D particle energy balance, which amounts to solving the following system of ODEs:

$$\begin{aligned}
\partial_t \rho_e \alpha_p^\star K_p &= -\rho_e \alpha_p^\star \frac{1}{\tau_c}(1 - e_c^2)\Theta_p \\
\partial_t \rho_e \alpha_p^\star e_p &= \rho_e \alpha_p^\star \frac{1}{\tau_c}(1 - e_c^2)\Theta_p
\end{aligned} \quad \Longrightarrow \quad \begin{aligned}
\partial_t \Theta_p &= -\frac{2}{3}\frac{1}{\tau_c}(1 - e_c^2)\Theta_p \\
\partial_t e_p &= -\frac{3}{2}\partial_t \Theta_p
\end{aligned} \tag{B.13}$$

with τ_c constant. The heating of particles through collision results in a transfer of granular temperature Θ_p to internal energy e_p. Conservation of particle energy states that $\rho_e \alpha_p^\star E_p$ is constant during collisional process. Note that in Houim and Oran (2016), the integration of the collisional source term takes into account the dependence of τ_c on Θ_p instead of considering τ_c constant during the collisional process.

The updated granular temperature is approximated by

$$\Theta_p(t) = \Theta_p(0) e^{-\frac{2}{3}\frac{(1-e_c^2)}{\tau_c}t} \tag{B.14}$$

The updated energies K_p and e_p are then retrieved from the updated granular temperature with $\rho_e \alpha_p^\star$ and u_p constant:

$$K_p(t) = \rho_e \alpha_p^\star \left(\frac{1}{2}u_p^2 + \frac{3}{2}\Theta_p(t) \right) \tag{B.15}$$

$$e_p(t) = \rho_e \alpha_p^\star \left(e_p(0) - \frac{3}{2}(\Theta_p(t) - \Theta_p(t)) \right) \tag{B.16}$$

For the frictional source term, the same methodology is used by replacing $\frac{1}{\tau_c}(1 - e_c^2)$ by $\frac{1}{\tau_{fr}}$.

Data availability

Data will be made available on request.

References

Batten, P., Clarke, N., Lambert, C., Causon, D.M., 1997. On the choice of wavespeeds for the HLLC Riemann solver. SIAM J. Sci. Comput. 18 (6), 1553–1570.
Behrendt, J., Balachandar, S., McGrath, T.P., 2022. Shock interacting with a random array of stationary particles underwater. Phys. Rev. Fluids 7 (2), 023401.
Benedetto, A. Di, Russo, P., Amyotte, P., Marchand, N., 2010. Modelling the effect of particle size on dust explosions. Chem. Eng. Sci. 65 (2), 772–779.
Benjelloun, S., Ghidaglia, J.-M., 2021. On the sound speed in two-fluid mixtures and the implications for CFD, model validation. Eur. J. Mech. B Fluids 90, 152–168.
Berzi, D., Jenkins, J.T., Richard, P., 2020. Extended kinetic theory for granular flow over and within an inclinded erodible bed. J. Fluid Mech. 885, A27.
Biesheuvel, A., Spoelstra, S., 1989. The added mass coefficient of a dispersion of spherical gas bubbles in liquid. Int. J. Multiph. Flow 15, 911–924.
Boniou, Victor, Fox, Rodney O., 2023. Shock–particle-curtain-interaction study with a hyperbolic two-fluid model: Effect of particle force models. Int. J. Multiph. Flow 169, 104591.
Boniou, Victor, Fox, Rodney O., Posey, Jacob W., Houim, Ryan W., 2024. A multiphase model for fluid–particle flows with added mass and phase change. Chem. Eng. J. 502, 157967.
Capecelatro, J., 2022. Modeling high-speed gas–particle flows relevant to spacecraft landings. Int. J. Multiph. Flow 150, 104008.
Chauchat, J., 2018. A comprehensive two-phase flow model for unidirectional sheet-flows. J. Hydraul. Res. 56, 15.
Chauchat, J., Médale, M., 2014. A three-dimensional numerical model for dense granular flows based on the $\mu(I)$ rheology. J. Comput. Phys. 256, 696–712.
Cook, T.L., Harlow, F.H., 1984. Virtual mass in multiphase flow. Int. J. Multiph. Flow 10 (6), 691–699.
Daniel, K.A., Wagner, J.L., 2022. The shock-induced dispersal of particle curtains with varying material density. Int. J. Multiph. Flow 152, 104082.
Davis, S.L., Dittmann, T.B., Jacobs, G.B., Don, W.S., 2013. Dispersion of a cloud of particles by a moving shock: Effects of the shape, angle of rotation, and aspect ratio. J. Appl. Mech. Tech. Phys. 54, 900–912.
DeMauro, E.P., Wagner, J.L., DeChant, L.J., Beresh, S.J., Turpin, A.M., 2019. Improved scaling laws for the shock-induced dispersal of a dense particle curtain. J. Fluid Mech. 876, 881–895.
Einfeldt, B., 1988. On Godunov-type methods for gas dynamics. SIAM J. Numer. Anal. 25 (2), 294–318.
Einfeldt, B., Munz, C.-D., Roe, P.L., Sjögreen, B., 1991. On Godunov-type methods near low densities. J. Comput. Phys. 92 (2), 273–295.
Fox, Rodney O., 2014. On multiphase turbulence models for collisional fluid–particle flows. J. Fluid Mech. 742, 368–424.
Fox, R.O., 2019. A kinetic-based hyperbolic two-fluid model for binary hard-sphere collisions. J. Fluid Mech. 877, 282–329.
Fox, R.O., 2021. Advanced Approaches in Turbulence: Theory, Modeling, Simulation, and Data Analysis for Turbulent Flows, Chapter Multiphase Turbulence. Elsevier, Amsterdam, Netherlands, pp. 307–372.
Fox, R.O., 2024. Recent advances in well-posed Eulerian models for polydisperse multiphase flows. Int. J. Multiph. Flow 172, 104715.
Fox, Rodney O., 2025. The particle–fluid–particle pressure tensor for ideal-fluid–particle flow. J. Fluid Mech. 1010, A8.
Fox, R.O., Laurent, F., Vié, A., 2020. A hyperbolic two-fluid model for compressible flows with arbitrary material-density ratios. J. Fluid Mech. 903, A5.
Fox, Rodney O., Posey, Jacob W., Houim, Ryan W., Laurent, Frédérique, 2024. A kinetic-based model for polydisperse, high-speed, fluid–particle flows. Int. J. Multiph. Flow 171, 104698.
Furfaro, D., Saurel, R., 2015. A simple HLLC-type Riemann solver for compressible non-equilibrium two-phase flows. Comput. & Fluids 111, 159–178.
Garzó, V., Dufty, J.W., Hrenya, C.M., 2007. Enskog theory for polydisperse granular mixtures. I. Navier–Stokes order transport. Phys. Rev. E 76 (3), 031303.
Gidaspow, D., 1994. Multiphase Flow and Fluidization: Continuum and Kinetic Theory Descriptions. Academic Press, New York, USA.
Gottlieb, S., Shu, C.-W., Tadmor, E., 2001. Strong stability-preserving high-order time discretization methods. SIAM Rev. 43 (1), 89–112.
Guillard, H., Nkonga, B., 2017. On the behaviour of upwind schemes in the low Mach number limit: A review. Handb. Numer. Anal. 18, 203–231.
Gunn, D.J., 1978. Transfer of heat or mass to particles in fixed and fluidized beds. Int. J. Heat Mass Transfer 21, 467–476.
Houim, R.W., Kuo, K.K., 2011. A low-dissipation and time-accurate method for compressible multi-component flow with variable specific heat ratios. J. Comput. Phys. 230 (23), 8527–8553.
Houim, R.W., Oran, E.S., 2016. A multiphase model for compressible granular–gaseous flows: formulation and initial tests. J. Fluid Mech. 789, 166–220.
Huang, K., Wu, H., Yu, H., Yan, D., 2011. Cures for numerical shock instability in HLLC solver. Internat. J. Numer. Methods Fluids 65 (9), 1026–1038.
Jenkins, J.T., Larcher, M., 2023. Dense, steady, fully developed fluid-particle flows over inclined, erodible beds. Phys. Rev. Fluids 8, 024303.
Jiang, G.-S., Shu, C.-W., 1996. Efficient implementation of weighted ENO, schemes. J. Comput. Phys. 126 (1), 202–228.
Johnson, P.C., Jackson, R., 1987. Frictional–collisional constitutive relations for granular materials, with application to plane shearing. J. Fluid Mech. 176, 67–93.
Kong, B., Fox, R.O., 2017. A solution algorithm for fluid–particle flows across all flow regimes. J. Comput. Phys. 344, 575–594.
Kong, B., Fox, R.O., 2020. A moment-based kinetic theory model for polydisperse gas–particle flow. Powder Technol. 365, 92–105.
Lhuillier, D., Chang, C.-H., Theofanous, T.G., 2013. On the quest for a hyperbolic effective-field model of disperse flows. J. Fluid Mech. 731, 184–194.
Ling, Y., Wagner, J.L., Beresh, S.J., Kearney, S.P., Balachandar, S., 2012. Interaction of a planar shock wave with a dense particle curtain: Modeling and experiments. Phys. Fluids 24 (11), 113301.
Liou, M.-S., 1996. A sequel to AUSM: AUSM$^+$. J. Comput. Phys. 129 (2), 364–382.
Liou, M.-S., 2006. A sequel to AUSM, Part II: AUSM$^+$-up for all speeds. J. Comput. Phys. 214 (1), 137–170.
Liou, M.-S., C.J. Steffen, Jr., 1993. A new flux splitting scheme. J. Comput. Phys. 107 (1), 23–39.
Lun, C.K.K., Savage, S.B., Jeffrey, D.J., Chepurniy, N., 1984. Kinetic theories for granular flow: inelastic particles in couette flow and slightly inelastic particles in a general flowfield. J. Fluid Mech. 140, 223–256.
Marchisio, D.L., Fox, R.O., 2013. Computational Models for Polydisperse Particulate and Multiphase Systems. Cambridge University Press, Cambridge, UK.
McGrath, T., Clair, J. St., Balachandar, S., 2018. Modeling compressible multiphase flows with dispersed particles in both dense and dilute regimes. Shock Waves 28, 533–544.
Mehrabadi, M., Tenneti, S., Garg, R., Subramaniam, S., 2015. Pseudo-turbulent gas-phase velocity fluctuations in homogeneous gas-solid flow: fixed particle assemblies and freely evolving suspensions. J. Fluid Mech. 770, 210–246.
Naumann, Z., Schiller, L., 1935. A drag coefficient correlation. Z. Vereins Dtsch. Ingenieure 77 (318), e323.
Passalacqua, A., Fox, R.O., 2011. Implementation of an interative solution procedure for multi-fluid gas–particle flow models on unstructured grids. Powder Technol. 213 (1), 174–187.
Passalacqua, A., Galvin, J.E., Vedula, P., Hrenya, C.M., Fox, R.O., 2011. A quadrature-based kinetic model for dilute non-isothermal granular flows. Commun. Comput. Phys. 10 (01), 216–252.
Pelanti, M., LeVeque, R.J., 2006. High-resolution finite volume methods for dusty gas jets and plumes. SIAM J. Sci. Comput. 28 (4), 1335–1360.
Richardson, J.F., Zaki, W.N., 1997. Sedimentation and fluidisation: Part I. Chem. Eng. Res. Des. 75, S82–S100.
Rogue, X., Rodriguez, G., Haas, J.F., Saurel, R., 1998. Experimental and numerical investigation of the shock-induced fluidization of a particles bed. Shock Waves 8 (1), 29–45.
Saurel, R., Abgrall, R., 1999. A simple method for compressible multifluid flows. SIAM J. Sci. Comput. 21 (3), 1115–1145.
Saurel, R., Favrie, N., Petitpas, F., Lallemand, M.-H., Gavrilyuk, S.L., 2010. Modelling dynamic and irreversible powder compaction. J. Fluid Mech. 664, 348–396.
Serna, S., Marquina, A., 2005. Capturing shock waves in inelastic granular gases. J. Comput. Phys. 209 (2), 787–795.
Shallcross, G.S., Fox, R.O., Capecelatro, J., 2020. A volume-filtered description of compressible particle-laden flows. Int. J. Multiph. Flow 122, 103138.
Spiteri, R.J., Ruuth, S.J., 2002. A new class of optimal high-order strong-stability-preserving time discretization methods. SIAM J. Numer. Anal. 40 (2), 469–491.
Sridhar, Archana, Fox, Rodney O., Capecelatro, Jesse, 2025. Turbulence transport in moderately dense gas–particle compressible flows. J. Fluid Mech. 1023, A35.
Tavanashad, V., Passalacqua, A., Fox, R.O., Subramaniam, S., 2019. Effect of density ratio on velocity fluctuations in dispersed multiphase flow from simulations of finite-size particles. Acta Mech. 230, 469–484.
Theofanous, T.G., Mitkin, V., Chang, C.-H., 2016. The dynamics of dense particle clouds subjected to shock waves. Part 1. Experiments and scaling laws. J. Fluid Mech. 792, 658–681.
Toro, E.F., Spruce, M., Speares, W., 1994. Restoration of the contact surface in the HLL-Riemann solver. Shock Waves 4 (1), 25–34.
Vié, A., Doisneau, F., Massot, M., 2015. On the anisotropic Gaussian velocity closure for inertial-particle laden flows. Commun. Comput. Phys. 17 (01), 1–46.
Wagner, J.L., Daniel, K., Downing, C., Grasser, T.W., Lynch, K.P., 2023. Shock–particle-curtain interactions at high Mach number. In: AIAA SCITECH 2023 Forum. p. 2303.
Zhang, F., Frost, D.L., Thibault, P.A., Murray, S.B., 2001. Explosive dispersal of solid particles. Shock Waves 10 (6), 431–443.